

Introduction To Algebraic Function Fields And Codes

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Abstract

The purpose of this project was to study an introduction algebraic function fields and codes from *Algebraic Function Fields And Codes* by Henning Stichtenoth [1]. The basic concepts and terms of algebraic function fields: Valuations and Places, Rational Function Field, Divisors, Reimann-Roch Space and Weil Differentials are explained. Their application in constructing algebraic geometry codes, which in particular have well defined bounds due to properties of function fields, is then explained.

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Chapter 1

Algebraic Function Fields

Definition 1.1 (Field). Let F is a nonempty set. Let $.$ and $+$ be binary operations on F . The system $(F, +, \cdot)$ is called a field if all of the following hold:

- (i) The additive operation is associative: $(x + y) + z = x + (y + z)$.
- (ii) F has a zero element 0 such that $x + 0 = x = 0 + x$.
- (iii) Every element $x \in F$ has a negative $-x \in F$ such that $x + (-x) = 0 = (-x) + x$.
- (iv) The additive operation is commutative: $x + y = y + x$.
- (v) The multiplicative operation is associative: $(xy)z = x(yz)$.
- (vi) F has $1 \neq 0$ such that $x1 = x = 1x$.
- (vii) Every element $0 \neq x \in F$ has an inverse $x^{-1} \in F$ such that $xx^{-1} = 1 = x^{-1}x$.
- (viii) The multiplicative operation is commutative: $xy = yx$.
- (ix) The multiplicative operation distributes over the additive operation: $x(y + z) = xy + xz$.

Definition 1.2. A mapping f between two fields $(F, +_1, *_1)$ and $(G, +_2, *_2)$ preserving operations, i.e.,

1. $f(a +_1 b) = f(a) +_2 f(b)$
2. $f(a *_1 b) = f(a) *_2 f(b)$.

is a homomorphism (similar to ring homomorphism). A 1-1 homomorphism is called an isomorphism.

Definition 1.3. If E is a subfield (i.e., a subset of F satisfying all properties of a field) F , then F is called an extension field of E or simply an extension of E . If F is an extension of E , then trivially F is a vector space over E . The dimension of F over E is usually written as $[F : E]$.

Definition 1.4. Let F be an extension of E . Then the dimension of F considered as a vector space over E is called the degree of F over E .

Therefore, the degree of F over E is written as $[F : E]$. If $[F : E] < \infty$, then F is called a finite extension of E (or simply finite over E) otherwise F is called an infinite extension of E .

Theorem 1.5. Let $E \subseteq F \subseteq K$ be fields. If $[K : F] < \infty$ and $[F : E] < \infty$, then (i) $[K : E] < \infty$. (ii) $[K : E] = [K : F][F : E]$.

Refer to [2] for proof.

Definition 1.6. An algebraic function field (or function field) F/K of one variable over a field K is a finite algebraic extension of $K(x)$ for some element $x \in F$ which is transcendental over K .

Definition 1.7. The set $\overline{K} := \{z \in F \mid z \text{ is algebraic over } K\}$ is the field of constants F/K . Note $\overline{K} \subseteq K \subsetneq F$. $F/\overline{K}$ is a function field over $\overline{K}$. We say that K is algebraically closed in F (or K is the full constant field of F) if $\overline{K} = K$.

Theorem 1.8. $z \in F$ is transcendental over K if and only if the extension $F/K(z)$ is of finite degree.

Proof. Let $z \in F$ which is an extension of $K(x)$, x transcendental over K . Hence $K \subset K(x) \subset F \cdots (1)$.

Let z is algebraic over $K \implies [K(z) : K] < \infty$. If $[F : K(z)] < \infty$, then

$[K(z) : K][F : K(z)] = [F : K] < \infty$, so $[K(x) : K] < \infty$ by (1), a contradiction as x is transcendental over K . Hence z is algebraic over $K \implies [K(z) : K]$ is of infinite degree. $\therefore [K(z) : K]$ is of finite degree $\implies z$ is transcendental. Conversely, assume that z is transcendental over K . We know that z is algebraic over $K(z) \implies [K(z) : K] < \infty$ and $[F : K] < \infty$ and $K \subset K(z) \subset F$. Hence $[F : K(z)][K(z) : K] = [F : K] \implies [F : K(z)] < \infty$. $\square$

An important example is the Rational Function Field, explained below. This is an important function field, will be studied later and has applications in Algebraic Geometry Codes.

Example 1.9. F/K is called rational function field if $F = K(x)$ for some $x \in F$ which is transcendental over K . Each element $0 \neq z \in K(x)$ has a unique representation:

$$z = a \cdot \prod_i p_i(x)^{n_i}.$$

Remark 1.10. F/K is often represented as a simple algebraic field extension of $K(x)$ i.e., $F = K(x, y)$ where $p(y) = 0$ for some irreducible polynomial $p(T) \in K(x)[T]$.

Chapter 2

Valuations and Places

Definition 2.1. A valuation ring of the function field F/K is a ring $O \subseteq F$ with the following properties:

1. $K \subsetneq O \subsetneq F$, and
2. for every $z \in F$ we have that $z \in O$ or $z^{-1} \in O$.

Example 2.2. In the case of a rational function field $K(x)$, let $p(x)$ be a monic polynomial $p(x) \in K[x]$, we consider the set

$$O_{p(x)} := \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], p(x) \nmid g(x) \right\}^1$$

is a valuation ring of $K(x)/K$, as:-

1. $K := \{z = \frac{f(x)}{g(x)} \mid g(x) = 1, f(x) = z \in K \subset K[x], p(x) \nmid 1\} \subsetneq O_{p(x)} \subsetneq K(x)$, as $K(x)$ is a field and will contain product and inverse of all elements.
2. as $K[x] \subset K(x)$ hence $O_{p(x)} \subsetneq K(x)$.

Remark 2.3. If $q(x)$ is another irreducible monic polynomial then $O_{p(x)} \neq O_{q(x)}$; as $p(x) \nmid q(x)$ and $q(x) \nmid p(x)$, hence $\frac{f(x)}{q(x)} \in O_{p(x)}$ but $\frac{f(x)}{q(x)} \notin O_{q(x)}$ and similarly for $\frac{f(x)}{p(x)}$.

¹defined as x is transcendental over K and denominator will never be zero

Proposition 2.4. *Let O be a valuation ring of the function field F/K . Then the following hold:*

- (a) *O is a local ring; i.e., O has a unique maximal ideal $P = O \setminus O^\times$, where $O^\times = \{z \in O \mid \text{there is an element } w \in O \text{ with } zw = 1\}$ is the group of units of O .*
- (b) *Let $0 \neq x \in F$. Then $x \in P \iff x^{-1} \notin O$.*
- (c) *For the field $\overline{K}$ of constants of F/K we have $\overline{K} \subseteq O$ and $\overline{K} \cap P = \{0\}$.*

Proof. (a) *Claim:* that $P := O \setminus O^\times$ is an ideal of O .

Proof of Claim:

1. Let $x \in P, z \in O$. Then $xz \notin O^\times$ (otherwise x would be a unit),- consequently $xz \in P$.
2. Let $x, y \in P$. w.l.o.g. let $x/y \in O$. Then $1 - x/y \in O$ and $x - y = y(1 - x/y) \in P$ by (1). Hence P is the unique maximal ideal of O .

This proves P is a unique maximal ideal of O , as any ideal Q such that $P \subsetneq Q$ contains a unit, hence $P = O$.

(b) $x \in P \iff x$ is non unit in $O \iff x^{-1} \notin O$.

(c) Let $z \in K$ and $z \notin O \implies z^{-1} \in O$.

As z^{-1} is algebraic over K , $\exists a_1, \dots, a_r \in K$ with $a_r(z^{-1})^r + \dots + a_1 z^{-1} + 1 = 0 \implies z = -(a_r(z^{-1})^{r-1} + \dots + a_1) \in O$. This is a contradiction to the assumption $z \notin O$. $\therefore \overline{K} \subseteq O$.

$K \cap P = \{0\}$ as K is set of constants(units).

□

Lemma 2.5. *Let O be a valuation ring of the algebraic function field F/K , let P be its maximal ideal and $0 \neq x \in P$. Let $x_1, \dots, x_n \in P$ be such that $x_1 = x$ and $x_i \in x_{i+1}P$ for $i = 1, \dots, n-1$. Then we have $n \leq [F : K(x)] < \infty$.*

Proof. From Remark 1.8, $F/K(x)$ is a finite extension. We shall show $x_1, \dots, x_n$ are L.I. (Linearly Independent) over $K(x)$. Let $\sum_{i=1}^n p_i(x)x_i = 0$ with $p_i(x) \in K(x)$. We can assume that all $p(x)$ are polynomials in x and that x does not

divide all of them. Let $a_i = p_i(0)$, the constant term of $p_i(x)$, and hence we can define $j \in 1, \dots, n$ by the condition $a_j \neq 0$ but $a_i = 0$ for all $i > j$. We obtain:

$$-p_j(x)x_j = \sum_{i \neq j} p_i(x)x_i \quad (2.1)$$

with $p_i(x) \in O$ for $i = 1, \dots, n$ ($asx = x1 \in P$), $x_i \in x_j P$ for $i < j$ and $p_i(x) = xg_i(x)$ for $i > j$, where $g_i(x)$ is a polynomial in x . Dividing (2.1) by x_j gives:-

$$p_j(x) = - \sum_{x_i}^{x_j} \frac{x_i}{x_j} p_i(x) - \sum_x^{x_j} \frac{x_i}{x_j} g_i(x) x_i \in P \quad (2.2)$$

Now $p_j(x) = a_j + xg_j(x) \implies a_j = p_j(x) - xg_j(x)$ and $xg_j(x) \in K[x] \subset O$ and $x \in P$, $a_j \in P \cap K = \{0\} \implies a_j = 0$, a contradiction. $\square$

Theorem 2.6. *Let O be a valuation ring of the function field F/K and let P be its unique maximal ideal. Then the following hold:*

- (a) P is a principal ideal.
- (b) If $P = tO$ then each $0 \neq z \in F$ has a unique representation of the form $z = t^n u$ for some $n \in \mathbb{Z}$ and $u \in O^\times$.
- (c) O is a principal ideal domain. More precisely, if $P = tO$ and $\{0\} \neq I \subseteq O$ is an ideal then $I = t^n O$ for some $n \in \mathbb{N}$.

Proof. (a) Let P is not principal, and choose $0 \neq x_1 \in P \iff x_1^{-1} \notin O$ (1). $\therefore P \neq x_1 O$ [for if $P = x_1 O$ then $y = x_1 O \in P$ has $x_1 \in P$ which would imply P is principal]. Also $P \not\subseteq x_1 O$ as P is maximal. Hence $\exists x_2 \in P \setminus x_1 O$ (2). By (1) and (2) we get $x_2 x_1^{-1} \notin O \iff x_1 x_2^{-1} \in P$ (by Proposition 2.4(b)) $\implies x_1 \in x_2 P$.

Similarly we can find $x_3 \in P \setminus x_2 P$ such that $x_3 \neq x_1$, for if $x_3 = x_1$, then $x_1 = x_3 \in x_2 P$, a contradiction.

Hence we can obtain an infinite chain $x_1, x_2, x_3, \dots$ in P such that $x_i \in x_{i+1} P \forall i \geq 1$, a contradiction to Lemma 2.5. Hence P is maximal.

(b) **Existence:-** As z or $z^{-1} \in O$ we can assume that $z \in O$. If $z \in O^\times$ then $z = t^0 z$. It remains to consider the case $z \in P$. There is a maximal

$m \geq 1$ with $z \in t^m O$, as the length of a sequence $x_1 = z, x_2 = t^{m-1}, x_3 = t^{m-2}, \dots, x_m = t$ is bounded by Lemma 2.1 as $P = tO \implies x_1 = z = t^m u \in x_2 O = t^{m-1} tO = t^m O \in$, similarly for others. Write $z = t^m u$ with $u \in O$. Then u must be a unit of O (otherwise $u \in P = tO$, so $u = tw$ with $w \in O$ and $z = t^{m+1} w \in t^{m+1} O$, a contradiction to the maximality of m). Hence $z = t^m O$.

Uniqueness of The Representation:- Let $z = t^n u$ with $u \in O^\times$. Let $z = t^m u_1 = t^n u_2 \implies t^{m-n} = u_2 u_1^{-1} \implies t$ is a unit, a contradiction as $t \in P$. Hence representation is unique.

(c) Let $0 \neq I \subseteq O$ be an ideal. The set $A := \{r \in \mathbb{N} | t^r \in I\}$ is non-empty (in fact, if $0 \neq x \in I$ then $x = t^r u$ with $u \in O^\times$ and hence $t^r = xu^{-1} \in I$). Put $n := \min(A)$, exists by Well Ordering Principle. We claim that $I = t^n O$. $I \supseteq t^n O$ as $t^n \in I$. Conversely let $0 \neq y \in I \implies y = t^s w$ with $w \in O^\times$, so $t^s \in I$ and $s \geq n$. It follows that $y = t^n \cdot t^{s-n} w \in t^n O \implies I \subseteq t^n O \implies I = t^n O$. Hence I is principal as $t^n \in I$ always. Hence O is PID. $\square$

Definition 2.7. 1. A place P of the function field F/K is the maximal ideal of some valuation ring O of F/K . Every element $t \in P$ such that $P = tO$ is called a prime element for P (other notations are local parameter or uniformizing variable).

2. $\mathbb{P}_F := \{P | P \text{ is a place of } F/K\}$.

If O is a valuation ring of F/K and P is its maximal ideal, then O is uniquely determined by P as $O = \{z \in F | z^{-1} \notin P\}$, by Proposition 2.4(b). Hence $O_P := O$ is called the valuation ring of the place P .

Definition 2.8. A discrete valuation of F/K is a function $v : F \rightarrow \mathbb{Z} \cup \infty$ with the following properties:

1. $v(x) = \infty \iff x = 0$
2. $v(xy) = v(x) + v(y)$ for all $x, y \in F$.
3. (triangle inequality) $v(x + y) \geq \min\{v(x), v(y)\}$ for all $x, y \in F$.
4. There exists an element $z \in F$ with $v(z) = 1$.
5. $v(a) = 0$ for all $0 \neq a \in K$.

Remark 2.9. 1. In this context the symbol ∞ means some element not in $\mathbb{Z}$ such that $\infty + \infty = \infty + n = n + \infty = \infty$ and $\infty > m$ for all $m, n \in \mathbb{Z}$.

2. $v : F \rightarrow \mathbb{Z} \cup \{\infty\}$ is onto.

Proof: From (1), (5) and (2.3) we get preimage of 1, 0 and ∞ . Let $n, m \in \mathbb{N}$. From (2) and (2.3) observe: $v(z^n) = n$. Let $n - m > 0$.
 $\implies n - m = v(z^{n-m}) = v(z^n) + v(z^{-m}) = n + v(z^{-m}) \implies v(z^{-m}) = -m$. Hence v is onto.

Remark 2.10. Let v be a discrete valuation of F/K . Fix a real number $0 < c < 1$ and define the function $|\cdot|_v : F \rightarrow \mathbb{R}$ by :

$$|z|_v := \begin{cases} c^{v(z)} & \text{if } z \neq 0, \\ 0 & \text{if } z = 0. \end{cases}$$

It is easily verified that this function has the properties of an ordinary absolute:

1. $|z|_v = c^{v(z)} \geq 0$.

2. $|z|_v = 0 \iff c^{v(z)} = 0 \iff v(z) = \infty \iff z = 0$.

3. Ordinary Triangle Inequality: $|x + y|_v \leq |x|_v + |y|_v$.

$v(x + y) \geq \min\{v(x), v(y)\} = v(x)$ (say) $\implies c^{v(x+y)} < c^{v(x)} \implies |x + y|_v \leq |x|_v \leq |x|_v + |y|_v$.

A stronger and very useful version of the Triangle Inequality can be derived from the:

Lemma 2.11 (Strict Triangle Inequality). Let v be a discrete valuation of F/K and let $x, y \in F$ with $v(x) \neq v(y)$. Then $v(x + y) = \min\{v(x), v(y)\}$.

Proof. Observe $v(ay) = v(y)$ for $0 \neq a \in K$ (by (2) and (5)) $\implies v(-y) = v(y)$. As $v(x) \neq v(y)$, let $v(x) < v(y)$. Let $v(x + y) \neq \min\{v(x), v(y)\}$, so $v(x + y) > v(x)$ by (2.3). Then we obtain $v(x) = v((x + y) - y) \geq \min\{v(x + y), v(y)\} > v(x)$, a contradiction. Hence $v(x + y) = \min\{v(x), v(y)\}$. $\square$

Definition 2.12. To a place $P \in \mathbb{P}_F$ we associate a function $v_P : F \rightarrow F \cup \{\infty\}$ that will prove to be a discrete valuation of F/K as follows: Choose a prime element t for P . Then every $0 \neq z \in F$ has a unique representation $z = t^n u$ with $u \in O_P^\times$ and $n \in \mathbb{Z}$. Define $v_P(z) := n$ and $v_P(0) := \infty$.

Observe that this definition depends only on P , not on the choice of t . Infact, if t' is another prime element for P then $P = tO = t'O$, so $t = t'w$ for some $w \in O^\times P$. Hence $t^n u = (t'^n w^n)u = t'^n (w^n u)$ with $w^n u \in O_P^\times$.

Theorem 2.13. *Let F/K be a function field. Then*

(a) *For a place $P \in \mathbb{P}_F$, the function v_P defined above is a discrete valuation of F/K . Moreover we have:-*

$$O_P = \{z \in F | v_P(z) \geq 0\}$$

$$O_P^\times = \{z \in F | v_P(z) = 0\}$$

$$P = \{z \in F | v_P(z) > 0\}$$

(b) *An element $x \in F$ is a prime element for P if and only if $v_P(x) = 1$.*

(c) *Conversely, let v is a discrete valuation of F/K . Then the set $P := \{z \in F | v(z) > 0\}$ is a place of F/K , and $O_P = \{z \in F | v(z) \geq 0\}$ is the corresponding valuation ring.*

(d) *Every valuation ring O of F/K is a maximal proper subring of F .*

Proof. (a) Properties (1)-(5) hold for v_P as:-

1. Clearly, $v(x) = \infty \iff x = 0$.

2. holds trivially.

3. (Triangle Inequality). Let $x, y \in F$ with $v_P(x) = n, v_P(y) = m$ and $n \leq m < \infty$. Hence $x = t^n u_1$ and $y = t^m u_2$ with $u_1, u_2 \in O_P^\times$.

$$\implies x + y = t^n (u_1 + t^{m-n} u_2) = t^n z \text{ with } z \in O_P.$$

If $z = 0$ we have $v_P(x+y) = \infty > \min\{n, m\}$, otherwise $z = t^k u$ with $k \geq 0$ and $u \in O_P^\times$ [by Theorem 2.6 (b)]. Hence $v_P(x+y) = v_P(t^{(n+k)} u) = n + k \geq n = \min\{v_P(x), v_P(y)\}$. Hence triangle inequality is proven.

4. $v(z) = 1$ if $t = z.u$.

5. Let $a \in K \implies a$ is a unit $\implies v(a) = 0$.

Other part:-

1. $z \in O_P$ can be written as $z = t^m u, m > 0$ by Theorem ???. Hence $O_P = \{z \in F | v_P(z) \geq 0\}$.
2. $z \in O_P^\times \implies z$ is a unit $\implies z = t^0 u \implies v_P(z) = 0 \implies O_P^\times = \{z \in F | v_P(z) = 0\}$.
3. $P = O_P \setminus O_P^\times \implies P = \{z \in F | v_P(z) > 0\}$

(b) If x is prime $\implies x \in P$ and $P = xO \implies x = x^t \cdot u, u \in O^\times \implies v_P(x) = 1$, for if $t > 1 \implies 1 = x^{t-1}u \in P$, a contradiction as 1 is a unit. Conversely let $v_P(x) = 1 \implies x = t \cdot u \implies xO = tuO = tO = P \implies x$ is prime.

(c) Trivial, from part (b) and Theorem 2.6(b).

(d) Let O be a valuation ring of F/K , $O \subsetneq O' \subsetneq F \setminus K$. Hence there exists $z \in F \setminus O$. Let $\langle z \rangle$ be an ideal of $O' \implies z^{-1} \notin O' \supseteq O$ else $F \subsetneq O' \implies F = O'$. Hence $z^{-1} \notin O$ and $z \notin O$ a contradiction as O is a valuation ring. Hence O is maximal. $\square$

Remark 2.14. 1. Let P be a place of F/K and let O_P be its valuation ring.

As P is a maximal ideal, the residue class ring O_P/P is a field. For $x \in O_P$ we define $x(P) \in O_P/P$ to be the residue class of x modulo P , for $x \in F \setminus O_P$ we put $x(P) := \infty$ (note that the symbol ∞ is used here in a different sense as in Definition 2.8).

2. By Proposition 2.4 we know that $K \subseteq O_P$ and $K \cap P = \{0\}$, so the residue class map $O_P \rightarrow O_P/P$ induces a canonical embedding of K into O_P/P as this is a ring homomorphism. Henceforth we shall always consider K as a subfield of O_P/P via this embedding. Observe that this argument also applies to $\bar{K}$ so we can consider $\bar{K}$ as a subfield of O_P/P as well.

Definition 2.15. Let $P \in \mathbb{P}_F$:-

- (a) $F_P := O_P/P$ is the residue class field of P defined as P is a maximal ideal of O_P . The map $x \rightarrow x(P)$ from F to $F_P \cup \{\infty\}$ is called the residue class map with respect to P . Sometimes we shall also use the notation $x + P := x(P)$ for $x \in O_P$.

(b) $\deg P := [F_P : K]$ is called the degree of P . A place of degree one is also called a **rational place** of F/K .

Proposition 2.16. *If P is a place of F/K and $0 \neq x \in P$ then*

$$\deg P \leq [F : K(x)] < \infty.$$

Proof. Observe that Theorem 1.8 $\implies [F : K(x)] < \infty$. Hence is sufficient to show that any elements $z_1, \dots, z_n \in O_P$, whose residue classes $z_1(P), \dots, z_n(P) \in F_P$ are L.I. over K , are L.I. over $K(x)$. Suppose $\exists$ a non-trivial linear combination

$$\sum_{i=1}^n p_i(x) z_i = 0, p_i(x) \in K(x).$$

W.l.o.g. we assume that $p_i(x)$ are polynomials in x , not all of them are divisible by x (this can be obtained by cancellation in linear combination above if all are divisible); i.e., $p_i(x) = a_i + x g_i(x)$ with $a_i \in K$ and $g_i(x) \in K[x]$, not all $a_i = 0$. As $x \in P$ and $g_i(x) \in O_P$, $p_i(x)(P) = a_i(P) = a_i$. Applying the residue class map to we obtain

$$0 = 0(P) = \sum_{i=1}^n p_i(x)(P) z_i(P) = a_i z_i(P).$$

This contradicts the linear independence of $z_1(P), \dots, z_n(P)$ over K . $\square$

Corollary 2.17. *The field $\overline{K}$ of constants of F/K is a finite field extension of K .*

Proof. We use the fact that $\mathbb{P}_F = p$ (which will be proved in Corollary 2.2). Choose some $P \in \mathbb{P}_F$. As $\overline{K}$ is embedded into F_P via the residue class map $O_P \rightarrow F_P$ as $\overline{K} \subseteq O_P$, it follows that $[\overline{K} : K] \leq [F_P : K] < \infty$. $\square$

Remark 2.18. *Let P be a rational place of F/K i.e., $\deg P = 1$. Then we have $F_P = K$, and the residue class map maps F to $K \cup \{\infty\}$. In particular, if K is an algebraically closed field, then all places are rational and we can read an element $z \in F$ as a function*

$$z : \begin{cases} \mathbb{P}_F \rightarrow K \cup \{\infty\} \\ P \rightarrow z(P) \end{cases} \quad (2.3)$$

This is why F/K is called a function field. The elements of K , interpreted as functions in the sense of (2.3), are constant functions. For this reason K is called the constant field of F .

Also the following terminology is justified by (2.3)

Definition 2.19. Let $z \in F$ and $P \in \mathbb{P}_F$. We say that P is a zero of z if $v_P(z) > 0$; P is a pole of z if $v_P(z) < 0$. If $v_P(z) = m > 0$, P is a zero of z of order m ; if $v_P(z) = -m < 0$, P is a pole of z of order m .

NOTE: To understand, z can be thought of as a meromorphic function and P as its zero or pole.

Theorem 2.20. Let F/K be a function field and let R be a subring of F with $K \subseteq R \subseteq F$. let $\{0\} \neq I \subsetneq R$ is a proper ideal of R . Then there is a place $P \in \mathbb{P}_F$ such that $I \subseteq P$ and $R \subseteq P$.

Proof. Consider the set

$$\mathcal{F} := \{S | S \text{ is a subring of } F \text{ with } R \subseteq S \text{ and } IS \neq S\}$$

(IS is by definition the set of all finite $\sum a_v s_v$ with $a_v \in I, s_v \in S$ it is an ideal of S). $\mathcal{F}$ is non-empty as $R \in \mathcal{F}$, and $\mathcal{F}$ is inductively ordered by inclusion. In fact, if $H \subseteq \mathcal{F}$ is a totally ordered subset of F then $T := \{S | S \in H\}$ is a subring of F with $R \subseteq T$. We show $IT \neq T$.

Suppose this is false, then $1 = \sum_1^n a_v s_v$ with $a_v \in I, s_v \in T$. [As $IT = T \implies ti = t'$ for some $t, t' \in T$ and $i \in I \implies 1 = t'^{-1}ti \in IT$]. As H is totally ordered there is an $S_0 \in H$ such that $s_1, \dots, s_n \in S_0$, so $1 = \sum_1^n a_v s_v \in IS_0$, a contradiction as it is a proper ideal.

By Zorn's lemma F contains a maximal element; i.e., there is a ring $O \subseteq F$ such that $R \subseteq O \subseteq F, IO \neq O$, and O is maximal with respect to these properties. We show that O is a valuation ring of F/K . As $I = \{0\}$ and $IO \neq O$ we have $O \subseteq F$ and $I \subseteq O \setminus O^\times$. Suppose there exists an element $z \in F$ with $z \notin O$ and $z^{-1} \notin O$. Then $IO[z] = O[z]$ and $IO[z^{-1}] = O[z^{-1}]$, and we can find $a_0, \dots, a_n, b_0, \dots, b_m \in IO$ with

$$1 = a_0 + a_1 z + \dots + a_n z^n \tag{2.4}$$

$$1 = b_0 + b_1 z^{-1} + \dots + b_m z^{-m}. \tag{2.5}$$

Clearly $n \geq 1$ and $m \geq 1$. We can assume that m, n in (2.4) and (2.5) are chosen minimally and $m \leq n$. We multiply (2.4) by $1 - b_0$ and (2.5) by $a_n z^n$ to obtain:-

$$1 - b_0 = (1 - b_0)a_0 + (1 - b_0)a_1z + \cdots + (1 - b_0)a_n z^n$$

and

$$0 = (b_0 - 1)a^n z^n + b_1 a^n z^{n-1} + \cdots + b_m a_n z^{n-m}$$

Adding these equations gives $1 = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1}$ with coefficients $c_i \in IO$. This is a contradiction to the minimality of n in (2.4). Hence we have proved that $z \in O$ or $z^{-1} \in O$ for all $z \in F$, hence O is a valuation ring of F/K . $\square$

Corollary 2.21. *Let F/K be a function field, $z \in F$ transcendental over K . Then z has at least one zero and one pole. In particular $\mathbb{P}_F \neq \emptyset$.*

Proof. Consider the ring $R = K[z]$ and the ideal $I = zK[z] \neq \{0\}$. By Theorem 2.3, there is a place $P \in \mathbb{P}_F$ with $z \in P$, hence P is a zero of z by Theorem 2.2. The same argument with $I = z^{-1}K[z] \neq \{0\}$ proves that z^{-1} has a zero $Q \in \mathbb{P}_F$. Then Q is a pole of z^{-1} . $\square$

We state a very useful result below.

Theorem 2.22 (Weak Approximation Theorem). *Let F/K be a function field, $P_1, \dots, P_n \in \mathbb{P}_F$ pairwise distinct places of F/K , $x_1, \dots, x_n \in F$ and $r_1, \dots, r_n \in \mathbb{Z}$. Then there is some $x \in F$ such that*

$$v_{P_i}(x - x_i) = r_i \text{ for } i = 1, \dots, n.$$

Proof. For simplicity we write v_i instead of v_{P_i} .

Step 1. There is some $u \in F$ with $v_1(u) > 0$ and $v_i(u) < 0$ for $i = 2, \dots, n$.

Proof of Step 1. We use mathematical induction. For $n = 2$ we observe that $O_{P_1} \subsetneq O_{P_2}$ and vice versa [since valuation rings are maximal proper subrings of F]. Therefore we can find $y_1 \in O_{P_1} \setminus O_{P_2}$ and $y_2 \in O_{P_2} \setminus O_{P_1}$.

$\implies v_1(y_1) \geq 0, v_2(y_1) < 0, v_1(y_2) < 0$ and $v_2(y_2) \geq 0$. and $u := y_1/y_2$ has the property $v_1(u) > 0, v_2(u) < 0$ as $v_i(1/x) = -v_i(x)$.

For $n > 2$ we have by induction hypothesis an element y with $v_1(y) >$

$0, v_2(y) < 0, \dots, v_{n-1}(y) < 0$. If $v_n(y) < 0$ the proof is finished.

If $v_n(y) \geq 0$, we choose z with $v_1(z) > 0, v_n(z) < 0$ and put $u := y + z^r$. Here $r \geq 1$ is chosen in such a manner that $r \cdot v_i(z) \neq v_i(y)$ for $i = 1, \dots, n-1$.
 $\implies v_1(u) \geq \min\{v_1(y), r \cdot v_1(z)\} > 0$ and $v_i(u) = \min\{v_i(y), r \cdot v_i(z)\} < 0$ for $i = 2, \dots, n$ (by Triangle and Strict Triangle Inequality respectively).

Step 2. There is some $w \in F$ such that $v_1(w-1) > r_1$ and $v_i(w) > r_i$ for $i = 2, \dots, n$.

Proof of Step 2. Choose u as in Step 1 and put $w := (1 + u^s)^{-1}$. We have, for sufficiently large $s \in \mathbb{N}$, $v_1(w-1) = v_1(-u^s(1 + u^s)^{-1}) = s \cdot v_1(u) > r_1$, and $v_i(w) = -v_i(1 + u^s) = -s \cdot v_i(u) > r_i$ for $i = 2, \dots, n$.

Step 3. Given $y_1, \dots, y_n \in F$, there is an element $z \in F$ with $v_i(z - y_i) > r_i$ for $i = 1, \dots, n$.

Proof of Step 3. Choose $s \in \mathbb{Z}$ such that $v_i(y_j) \geq s$ for all $i, j \in 1, \dots, n$. By Step 2 there are $w_1, \dots, w_n$ with

$$v_i(w_i - 1) > r_i - s \text{ and } v_i(w_j) > r_i - s \text{ for } j \neq i.$$

Then $z := \sum_{j=1}^n y_j w_j \implies v_i(z - y_i) > \min \{s + r_i - s, \dots, s + r_i - s\} \text{ (n times)} = r_i$ by Triangle Inequality.

Now, by Step 2 we can find $z \in F$ with $v_i(z - x_i) > r_i, i = 1, \dots, n$:

Choose z_i with $v_i(z_i) = r_i$. By Step 3 there is z' with $v_i(z' - z_i) > r_i$ for $i = 1, \dots, n$.

$$\implies v_i(z) = v_i((z - z_i) + z_i) = \min \{v_i(z - z_i), v_i(z_i)\} = r_i.$$

Let $x := z + z'$, then: $v_i(x - x_i) = v_i((z - x_i) + z) = \min\{v_i(z - x_i), v_i(z)\} = r_i$. □

Corollary 2.23. *Every function field has infinitely many places.*

Proof. Suppose there are only finitely many places, say $P_1, \dots, P_n$. By Weak Approximation Theorem 2.22 $\exists x \in F$ with $v_{P_i}(x) > 0$ for $i = 1, \dots, n$. Hence $x \in P_i \forall i \implies$ all places of z are zeros. Hence z has no pole. Hence z has infinitely many places. □

Proposition 2.24. *Let F/K be a function field and let $P_1, \dots, P_r$ be zeros of the element $x \in F$. Then*

$$\sum_{i=1}^r v_{P_i(x)} \cdot \deg P_i \leq [F : K(x)].$$

Proof. Notations:

1. $v_i := v_{P_i}$
2. $f_i := \deg P_i$
3. $e_i := v_i(x)$.

$\forall i \exists t_i$ such that $v_i(t_i) = 1$ and $v_k(t_i) = 0$ for $k \neq i$. Next we choose $s_{i_1}, \dots, s_{i_{f_i}} \in O_{P_i}$ such that $s_{i_1}(P_i), \dots, s_{i_{f_i}}(P_i)$ form a basis of the residue class field F_{P_i} over K . By a weak application of Theorem 2.22 we can find $z_{ij} \in F$ such that the following holds for all i, j :

$$v_i(s_{ij} - z_{ij}) > 0 \text{ and } v_k(z_{ij}) > e_k \text{ for } k \neq i. \quad (2.6)$$

Claim: that the elements

$$t_i^a \cdot z_{ij}, 1 \leq i \leq r, 1 \leq j \leq f_i, 0 \leq a < e_i$$

are L.I. over $K(x)$ hence contained in basis for $F/K(x)$. Their number is equal to $\sum_{i=1}^r f_i e_i = \sum_{i=1}^r v_{P_i(x)} \cdot \deg P_i$ hence the Proposition follows. Suppose there is a non-trivial linear combination

$$\sum_{i=1}^r \sum_{j=1}^{f_i} \sum_{a=0}^{e_i-1} \psi_{ija}(x) t_i^a z_{ij} = 0 \quad (2.7)$$

over $K(x)$. W.l.o.g. we can assume that $\psi_{ija}(x) \in K[x]$ and not all $\psi_{ija}(x)$ are divisible by x . Then there are indices $k \in 1, \dots, r$ and $c \in 0, \dots, e_k - 1$ such that

$$x | \psi_{kja}(x) \text{ for all } a < c \text{ and all } j \in \{1, \dots, f_k\}, \text{ and } x \nmid \psi_{kjc}(x) \text{ for some } j \in \{1, \dots, f_k\}. \quad (2.8)$$

Multiplying (2.7) by t_k^{-c} we obtain:-

$$\sum_{i=1}^r \sum_{j=1}^{e_i-1} \sum_{a=0}^{e_i-1} \psi_{ija}(x) t_i^a t_k^{-c} z_{ij} = 0 \quad (2.9)$$

For $i \neq k$ all summands are in P_k as

$$v_k(\psi_{ija}(x) t_i^a t_k^{-c} z_{ij}) = v_k(\psi_{ija}(x)) + av_k(t_i) - cv_k(t_k) + v_k(z_{ij}) \geq 0 + 0 - c + e_k > 0$$

(Note that $x | \psi_{kja}(x)$ and hence $v_k(\psi_{kja}(x)) = v_k(f(x)) + v_k(x) \geq e_k$ where $f(x) = \psi_{kja}(x)/x$. for $i = k$ and $a > c$,

$$v_k(\psi_{kja}(x) t_k^{a-c} z_{kj}) \geq a - c > 0.$$

Combining the above we get:-

$$\implies \sum_{j=1}^{f_k} \psi_{kjc}(x) z_{kj} \in P_k \quad (2.10)$$

Not all $\psi_{kjc} P_i = 0$ by (4.3), as $x \nmid$ all ψ . Hence (2.10) gives non trivial linear combination:-

$$\sum_{j=1}^{f_k} \psi_{kjc}(x) (P_k) \cdot z_{kj}(P_k) = 0$$

a contradiction as $z_{k1}(P_k), \dots, z_{kf_i}(P_k)$ forms a basis of F_{P_k}/K . □

Corollary 2.25. *In a function field F/K every element $0 \neq x \in F$ has only finite number of zeros or poles.*

Proof. If x is constant, x has neither zeros nor poles. If x is transcendental over K , the number of zeros is $\leq [F : K(x)]$ by Proposition 2.24. The same argument shows that x^{-1} has only a finite number of zeros. □

Chapter 3

The Rational Function Field

We now study what the concepts of previous section in case of the rational function field $F = K(x)$, where x is transcendental over K

Given an irreducible monic polynomial $p(x) \in K[x]$ we consider the valuation ring:-

$$O_{p(x)} := \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], p(x) \nmid g(x) \right\} \quad (3.1)$$

of $K(x)/K$ with maximal ideal

$$P_{p(x)} = \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], p(x) \mid f(x), p(x) \nmid g(x) \right\} \quad (3.2)$$

as $O_{p(x)}^\times = \left\{ h(x) = \frac{f(x)}{g(x)} \mid p(x) \nmid f(x), p(x) \nmid g(x) \right\}$ is set of units of $K(x)$ as $1/h(x)$ (inverse) is defined here only and maximal ideal $P = O \setminus O^\times$.

In the particular case when $p(x)$ is linear, i.e. $p(x) = x - \alpha$ with $\alpha \in K$, we abbreviate and write:-

$$P_\alpha := P_{x-\alpha} \in \mathbb{P}_K(x). \quad (3.3)$$

There is another valuation ring of $K(x)/K$, namely

$$O_\infty := \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], \deg f(x) \leq \deg g(x) \right\} \quad (3.4)$$

with maximal ideal

$$P_\infty = \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], \deg f(x) < \deg g(x) \right\} \quad (3.5)$$

as set of units is $O_P^\times = \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], \deg f(x) = \deg g(x) \right\}$ for if $\deg f(x) \neq \deg g(x)$ then inverse $\notin O_\infty$. This place is called the infinite place of $K(x)$.

Remark 3.1. Observe that these labels depend on the specific choice of the generating element x of $K(x)/K$. For example $K(x) = K(1/x)$, and the infinite place with respect to $1/x$ is the place P_0 with respect to x as:-

$$\begin{aligned} P_\infty(1/x) &= \left\{ \frac{f(1/x)}{g(1/x)} \mid f(1/x), g(1/x) \in K[1/x], \deg f(1/x) < \deg g(1/x) \right\} \\ &= \left\{ \frac{a_n(1/x)^n + a_{n-1}(1/x)^{n-1} + \cdots + a_1(1/x) + a_0}{b_m(1/x)^m + b_{m-1}(1/x)^{m-1} + \cdots + b_1(1/x) + b_0} \mid m < n \right\} \\ &= \left\{ \frac{a_n + a_{n-1}(x)^1 + \cdots + a_1(x)^{n-1} + a_0^n}{b_m(x)^{n-m} + b_{m-1}(x)^{n-m+1} + \cdots + b_1(x)^{n-1} + b_0x^n} \mid m < n \right\} \\ &= \left\{ \frac{F(x)}{G(x)} \mid F(x), G(x) \in K[x], x \nmid F(x), x \nmid G(x) \right\} = P_0(x) \end{aligned}$$

Proposition 3.2. Let $F = K(x)$ be the rational function field.

- (a) Let $P = P_{p(x)} \in \mathbb{P}_{K(x)}$ be the place defined by (3.2), where $p(x) \in K[x]$ is an irreducible polynomial. Then $p(x)$ is a prime element for P , and the corresponding valuation v_P can be described as follows: if $z \in K(x) \setminus 0$ is written in the form $z = p(x)^n(f(x)/g(x))$ with $n \in \mathbb{Z}$, $f(x), g(x) \in K[x]$, $p(x) \nmid f(x)$ and $p(x) \nmid g(x)$, then $v_P(z) = n$. The residue class field $K(x)_P = O_P/P$ is isomorphic to $K[x]/(p(x))$ an isomorphism is given by

$$p: \begin{cases} K[x]/(p(x)) \rightarrow K(x)_P, \\ f(x) \bmod p(x) \mapsto f(x)(P). \end{cases}$$

Consequently $\deg P = \deg p(x)$.

- (b) In the special case $p(x) = x - \alpha$ with $\alpha \in K$ the degree of $P = P_\alpha$ is one, and the residue class map is given by

$$z(P) = z(\alpha) \text{ for } z \in K(x),$$

where $z(\alpha)$ is defined as follows: write $z = f(x)/g(x)$ with relatively prime polynomials $f(x), g(x) \in K[x]$. Then

$$z(\alpha) = \begin{cases} f(\alpha)/g(\alpha) & \text{if } g(\alpha) \neq 0, \\ \infty & \text{if } g(\alpha) = 0. \end{cases}$$

(c) Finally, let $P = P_\infty$ be the infinite place of $K(x)/K$ defined by (3.5). Then $\deg P_\infty = 1$. A prime element for P_∞ is $t = 1/x$. The corresponding discrete valuation v_∞ is given by

$$v_\infty(f(x)/g(x)) = \deg g(x) - \deg f(x),$$

where $f(x), g(x) \in K[x]$. The residue class map corresponding to P_∞ is determined by $z(P_\infty) = z(\infty)$ for $z \in K(x)$, where $z(\infty)$ is defined as usual: if

$$z = \frac{a^n x^n + \cdots + a_0}{b^m x^m + \cdots + b_0} \text{ with } a_n, b_m \neq 0,$$

then

$$z(\infty) = \begin{cases} a_n/b_n & \text{if } n = m, \\ 0 & \text{if } n < m, \\ \infty & \text{if } n > m. \end{cases}$$

(d) K is the full constant field of $K(x)/K$.

Proof. (a)

$$P_{p(x)} = p(x)O_{p(x)}$$

clearly from definition and hence the valuation v_P is defined. Now consider the ring homomorphism

$$\psi : \begin{cases} K[x] \rightarrow K(x)_P, \\ f(x) \mapsto f(x)(P). \end{cases}$$

1. Finding $\text{Ker } \psi$:

$$\psi(f(x)) = f(x) + P_{p(x)} = P_{p(x)} \implies p(x) | f(x) \implies \text{Ker } \psi = (p(x)).$$

2. Showing ψ is onto:

Let $z \in O_{p(x)} \implies z = u(x)/v(x)$ with $u(x), v(x) \in K[x]$ such that

$p(x) \nmid v(x)$. Hence there are $a(x), b(x) \in K[x]$ with $a(x)p(x) + b(x)v(x) = 1$ hence

$$z = 1. z = \frac{a(x)u(x)}{v(x)}p(x) + b(x)u(x),$$

hence $z(P) = (b(x)u(x))(P) \in K(x)_P$ with $(b(x)u(x)) \in K[x]$.

Hence by Fundamental Theorem of Ring Isomorphism we get:-

$$\frac{K[x]}{(p(x))} \cong K(x)_P.$$

Hence $\deg P = \deg \frac{K[x]}{(p(x))} = \deg(p(x))$.

(b) Now $P = P_\alpha$ with $\alpha \in K$. If $f(x) \in K[x]$ then $(x - \alpha) \mid (f(x) - f(\alpha))$ hence $f(x)P = (f(x) - f(\alpha))P - f(\alpha)P = f(\alpha)$. An arbitrary element $z \in O_P$ can be written as $z = f(x)/g(x)$ with polynomials $f(x), g(x) \in K[x]$ and $(x - \alpha) \nmid g(x)$, hence $g(x)(P) = g(\alpha) \neq 0$ and $z(P) = \frac{f(x)(P)}{g(x)(P)} = \frac{f(\alpha)}{g(\alpha)} = z(\alpha)$.

(c) $\deg P = \deg p(x) = 1$. We will only show that $1/x$ is a prime element for P_∞ . We have $1/x \in P$. Let $z = f(x)/g(x) \in P_\infty$; i.e., $\deg f < \deg g$. Then $z = \frac{1}{x} \frac{xf}{g}$ with $\deg(xf) \neq \deg g \implies z \in (1/x)O_\infty$ hence $1/x$ is prime element P_∞ .

(d) Choose a place P of $K(x)/K$ of degree one (e.g. $P = P_\alpha$ with $\alpha \in K$). The field $\overline{K}$ of constants of $K(x)$ is embedded into the residue class field $K(x)_P$ by Remark 2.6(2), hence $K \subseteq \overline{K} \subseteq K(x)_P = K$. $\square$

Theorem 3.3. *There are no places of the rational function field $K(x)/K$ other than the places $P_p(x)$ and P_∞ .*

Proof. Let P be a place of $K(x)/K$. We distinguish two cases as follows:

Case 1. Assume that $x \in O_P$. Then $K[x] \subseteq O_P$. Set $I := K[x] \cap P \neq \{0\}$ by Proposition 2.4. this is a prime ideal $K[x]$ as every element is a polynomial in x . The residue class map induces an embedding $K[x]/I \hookrightarrow K(x)_P$. It follows that there is a (uniquely determined) irreducible monic polynomial $p(x) \in K[x]$ such that $I = K[x] \cap P = p(x).K[x]$. Every $g(x) \in K[x]$ with $p(x) \nmid g(x)$ is not in I , so $g(x) \notin P$ and $1/g(x) \in O_P$ by Proposition 2.4(b). Hence:-

$O_{p(x)} = \left\{ \frac{f(x)}{g(x)} \mid f(x), g(x) \in K[x], p(x) \nmid g(x) \right\} \subseteq O_P$. As valuation rings are maximal proper subrings of $K(x)$ by Theorem 2.6(d) hence $O_P = O_{p(x)}$.

Case 2. Now $x \in O_P$. We conclude that $K[x^{-1}] \subseteq O_P, x^P - 1 \in P \cap K[x^{-1}]$ and $P \cup K[x^{-1}] = x^{-1}K[x^{-1}]$.

$$\begin{aligned} \text{As in case 1, } O_P &\supseteq \left\{ \frac{f(x^{-1})}{g(x^{-1})} \mid f(x^{-1}), g(x^{-1}) \in K[x^{-1}], x^{-1} \nmid g(x^{-1}) \right\} \\ &= \left\{ \frac{a_0 + a_1 x^{-1} + \dots + a_n x^{-n}}{b_0 + b_1 x^{-1} + \dots + b_m x^{-m}} \mid b_0 \neq 0 \right\} = \left\{ \frac{a_0 x^{m+n} + \dots + a_n x_m}{b_0 x^{m+n} + \dots + b_m x^n} \mid b_0 \neq 0 \right\} \\ &= \left\{ \frac{u(x)}{v(x)} \mid u(x), v(x) \in K[x], \deg u(x) \leq \deg v(x) \right\} = O_\infty. \end{aligned}$$

Hence $O_P = O_\infty$ and $P = P_\infty$. □

Corollary 3.4. *The places of $K(x)/K$ of degree one are in 1-1-correspondence with $K \cup \{\infty\}$.*

Proof. By previous theorem, the only possible places of degree 1 are $P_{x-\alpha}, \alpha \in K$ and P_∞ . Also $\deg P_{x-\alpha} = 1$ by Theorem 3.2(a). Note that Remark 3.1 $P_\infty(x) = P_0(1/x)$ which is of degree 2. Hence $\deg P_\infty(x) = 1$. Clearly we can find a 1-1-correspondence with $K \cup \{\infty\}$. □

Chapter 4

Divisors

NOTE: From here on, F/K always denotes an algebraic function field of one variable such that $K = \overline{K}$ is the full constant field of F/K .

Definition 4.1. *The divisor group $\text{Div}(F)$ of F/K is defined as the (additively written) free abelian group which is generated by the places of F/K . The elements of $\text{Div}(F)$ are called divisors of F/K . In other words, a divisor is a formal sum*

$$D = \sum_P n_P P \text{ with } n_P \in \mathbb{Z}, \text{ almost all } n_P = 0.$$

The support of D is defined as

$$\text{supp } D := \{P \in \mathbb{P}_F \mid n_P \neq 0\}.$$

It will often be found convenient to write

$$D = \sum_{P \in S} n_P P$$

where $S \subseteq \mathbb{P}_F$ is a finite set with $S \supseteq \text{supp } D$. A divisor of the form $D = P$ with $P \in \mathbb{P}_F$ is called a prime divisor. Two divisors $D = \sum n_P P$ and $D' = \sum n'_P P$ are added coefficientwise.

$$D + D' = \sum_{P \in \mathbb{P}_F} (n_P + n'_P) P$$

The zero element of the divisor group $\text{Div}(F)$ is the divisor

$$0 := \sum_{P \in \mathbb{P}_F} r_P P, \text{ all } r_P = 0.$$

For $Q \in \mathbb{P}_F$ and $D = n_P P \in \text{Div}(F)$ we define $v_Q(D) := n_Q$, hence $\text{supp } D = \{P \in \mathbb{P}_F \mid v_P(D) > 0\}$ and $D = \sum_{P \in \text{supp } D} v_P(D) \cdot P$. A partial ordering on $\text{Div}(F)$ is defined by

$$D_1 \leq D_2 : \iff v_P(D_1) \leq v_P(D_2) \text{ for all } P \in \mathbb{P}_F.$$

If $D_1 \leq D_2$ and $D_1 \neq D_2$. We will also write $D_1 < D_2$. A divisor $D \geq 0$ is called positive (or effective).

The degree of a divisor is defined as $\deg D := \sum_{P \in \mathbb{P}_F} v_P(D) \cdot \deg P$, and this gives a homomorphism $\deg: \text{Div}(F) \rightarrow \mathbb{Z}$ as $\deg(D_1 + D_2) := \sum_{P \in \mathbb{P}_F} (v_P(D_1) + v_P(D_2)) \cdot \deg P$.

A nonzero element $x \in F$ has only finitely many zeros and poles in $\mathbb{P}_F$ by Corollary 2.24.

Hence the following definition makes sense.

Definition 4.2. Let $0 \neq x \in F$ and denote by Z (resp. N) the set of zeros (resp. poles) of x in $\mathbb{P}_F$. Then we define

$$(x)_0 := \sum_{P \in Z} v_P(x) P, \text{ the zero divisor of } x$$

$$(x)_\infty := \sum_{P \in N} (v_P(x)) P, \text{ the pole divisor of } x$$

$$(x) := (x)_0 - (x)_\infty \text{ the principal divisor of } x.$$

Clearly $(x)_0 \geq 0$, $(x)_\infty \geq 0$ and

$$(x) = \sum_{P \in \mathbb{P}_F} v_P(x) P. \tag{4.1}$$

The elements $0 \neq x \in F$ which are constant are characterized by $x \in K \iff (x) = 0$ (note the general assumption made previously that K is algebraically closed in F).

Definition 4.3. *The set of divisors*

$$\text{Princ}(F) := \{(x) \mid 0 \neq x \in F\}$$

is called the group of principal divisors of F/K . This is a subgroup of $\text{Div}(F)$, as for $0 \neq x, y \in F$, $(xy) = (x) + (y)$ by (5.1). The factor group

$$\text{Cl}(F) := \text{Div}(F)/\text{Princ}(F)$$

is called the divisor class group of F/K . For a divisor $D \in \text{Div}(F)$, the corresponding element in the factor group $\text{Cl}(F)$ is denoted by $[D]$, the divisor class of D . Two divisors $D, D' \in \text{Div}(F)$ are said to be equivalent, written $D \sim D'$ if $[D] = [D']$ i.e., $D' = D + (x)$ for some $x \in F \setminus \{0\}$. This is easily verified to be an equivalence relation.

Chapter 5

Reimann-Roch Space

Definition 5.1. For a divisor $A \in \text{Div}(F)$ we define the Riemann-Roch space associated to A by

$$\mathcal{L}(A) := \{x \in F \mid (x) \geq -A\} \cup \{0\}.$$

This definition has the following interpretation: if

$$A = \sum_{i=1}^r n_i P_i - \sum_{j=1}^s m_j Q_j$$

with $n_i > 0, m_j > 0$ then $L(A)$ consists of all elements $x \in F$ such that

- x has zeros of order $\geq m_j$ at Q_j , for $j = 1, \dots, s$, and
- x may have poles only at the places $P_1, \dots, P_r$ with the pole order at P_i being bounded by n_i ($i = 1, \dots, r$).

Remark 5.2. Let $A \in \text{Div}(F)$. Then

1. $x \in \mathcal{L}(A)$ if and only if $v_P(x) \geq -v_P(A)$ for all $P \in \mathbb{P}_F$.
2. $L(A) \neq \{0\}$ if and only if there is a divisor $A \sim A'$ with $A' \geq 0$.

Lemma 5.3. Let $A \in \text{Div}(F)$. Then we have:

1. $\mathcal{L}(A)$ is a vector space over K .

2. If A is a divisor equivalent to A , then $\mathcal{L}(A) \cong \mathcal{L}(A')$ (isomorphic as vector spaces over K).

Proof. (a) Let $x, y \in L(A)$ and $a \in K$. Then for all $P \in \mathbb{P}_F$, $v_P(x + y) \geq \min\{v_P(x), v_P(y)\} \geq -v_P(A)$ and $v_P(ax) = v_P(a) + v_P(x) \geq -v_P(A)$. So $x + y$ and ax are in $\mathcal{L}(A)$.

(b) By assumption, $A = A + (z)$ with $0 = z \in F$. Consider the mapping

$$p : \begin{cases} \mathcal{L}(A) \mapsto F \\ x \rightarrow xz \end{cases}$$

This is a K -linear mapping whose image is contained in $\mathcal{L}(A)$. In the same manner,

$$p : \begin{cases} \mathcal{L}(A') \rightarrow F \\ x \rightarrow xz^{-1} \end{cases}$$

is K -linear from $\mathcal{L}(A)$ to $\mathcal{L}(A')$. These mappings are inverse to each other, hence p is an isomorphism between $\mathcal{L}(A)$ and $\mathcal{L}(A')$. $\square$

Lemma 5.4. (a) $L(0) = K$.

(b) If $A < 0$ then $\mathcal{L}(A) = \{0\}$.

Proof. (a) We have $(x) = 0$ for $0 \neq x \in K$ is a unit and hence has no poles or zeros, hence $K \subseteq L(0)$. Conversely, if $0 \neq x \in L(0)$ then $(x) \geq 0$. This means that x has no pole, so $x \in K$ by Corollary .

(b) Assume there exists an element $0 \neq x \in \mathcal{L}(A)$. Then $(x) \geq -A > 0$ which implies that x has at least one zero but no pole. This is impossible. $\square$

Lemma 5.5. Let A, B be divisors of F/K with $A \geq B$. Then we have $\mathcal{L}(A) \subseteq \mathcal{L}(B)$ and $\dim(\mathcal{L}(B)/\mathcal{L}(A)) \leq \deg B - \deg A$.

Proof. $L(A) \subseteq L(B)$ is trivial. In order to prove the other assertion we can assume that $B = A + P$ for some $P \in \mathbb{P}_F$; the general case follows then by induction. Choose an element $t \in F$ with $v_P(t) = v_P(B) = v_P(A) + 1 \cdots (**)$. For $x \in L(B)$ we have $v_P(x) \leq -v_P(B) = -v_P(t)$, so $xt \in O_P$. Hence we obtain a K -linear map:-

$$\psi : \begin{cases} L(B) \rightarrow F_P \\ x \mapsto (xt)(P) \end{cases}$$

Now $x \in \text{Ker } \psi \iff xt \in P \iff v_P(xt) > 0$; i.e., $v_P(x) \geq -v_P(A)$ using (**)
 $\iff x \in \mathcal{L}(A)$, Hence $\text{Ker } \psi = \mathcal{L}(A)$ induces a K -linear injective mapping
from $\mathcal{L}(B)/\mathcal{L}(A)$ to F_P . It follows that $\dim(\mathcal{L}(B)/\mathcal{L}(A)) \leq \dim F_P =$
 $\deg B - \deg A$. $\square$

Proposition 5.6. *For each divisor $A \in \text{Div}(F)$ the space $\mathcal{L}(A)$ is a finite dimensional vector space over K . More precisely: if $A = A_+ - A_-$ with positive divisors A_+ and A_- , then*

$$\dim \mathcal{L}(A) \leq \deg A_+ + 1.$$

Proof. As $\mathcal{L}(A) \subseteq \mathcal{L}(A_+)$, it is sufficient to show that $\dim \mathcal{L}(A_+) \geq \deg A_+ + 1$. We have $0 \geq A_+$, so Lemma 5.5 gives $\dim(\mathcal{L}(A_+)/\mathcal{L}(0)) \leq \deg A_+$. As $\mathcal{L}(0) = K$ we conclude that

$$\dim \mathcal{L}(A_-) = \dim(\mathcal{L}(A_+)/\mathcal{L}(0)) + 1 \geq \deg A_+ + 1.$$

$\square$

Theorem 5.7. *All principal divisors have degree zero. More precisely: let $x \in F \setminus K$ and $(x)_0$ resp. $(x)_\infty$ denote the zero resp. pole divisor of x . Then $\deg (x)_0 = \deg (x)_\infty = [F : K(x)]$.*

Proof. Set $n := [F : K(x)]$ and $B := (x)_\infty = \sum_{i=1}^r -v_{P_i}(x)P_i$ where $P_1, \dots, P_r$ are all the poles of x .

$$\implies \deg B = \sum_{i=1}^r v_{P_i}(x^{-1}) \deg P_i \leq [F : K(x)] = n \text{ by Proposition 2.24}$$

It remains to show that $n \leq \deg B$. Choose a basis $u_1, \dots, u_n$ of $F/K(x)$ and a divisor $C \geq 0$ such that $(u_i) \geq -C$ for $i = 1, \dots, n$. We have

$$\ell(lB + C) \geq n(l + 1) \quad \forall l \geq 0 \tag{5.1}$$

as $x_i u_j \in \mathcal{L}(lB + C)$ for $0 \leq i \leq l, 1 \leq j \leq n$ [$u_1, \dots, u_n$ are L.I. over $K(x)$]. Setting $c := \deg C$ we obtain $n(l + 1) \leq (lB + C) \leq l \deg B + c + 1$.

$$\implies l(\deg B - n) \geq n - c - 1 \quad \forall l \in \mathbb{N}. \tag{5.2}$$

The right hand side of (5.2) is independent of l , possible only when $\deg B \geq n$. We have hence proved that $\deg(x)_\infty = [F : K(x)]$. As $(x)_0 = (x^{-1})_\infty \implies \deg(x)_0 = \deg(x^{-1})_\infty = [F : K(x^{-1})] = [F : K(x)]$. $\square$

Definition 5.8. For $A \in \text{Div}(F)$ the integer $\ell(A) := \dim \mathcal{L}(A)$ is called the dimension of the divisor A .

Corollary 5.9. (a) Let A, A' be divisors with $A \sim A'$. Then we have $\ell(A) = \ell(A')$ and $\deg A = \deg A'$.

(b) If $\deg A < 0$ then $\ell(A) = 0$.

(c) For a divisor A of degree zero the following assertions are equivalent:

1. A is principal.
2. $\ell(A) \geq 1$.
3. $\ell(A) = 1$.

Proof. (a) $A \sim A' \implies \mathcal{L}(A) \cong \mathcal{L}(A') \implies \ell(A) = \ell(A')$.

(b) Let $\deg A < 0$, by Remark 5.2(b) there is some divisor $A \sim A'$ with $\deg A' < 0$, hence $\deg A = \deg A' < 0$.

(c) (1) $\implies$ (2): If $A = (x)$ is principal then $x^{-1} \in \mathcal{L}(A)$ as $v_P(x^{-1}) = -v_P(x)$, so $\ell(A) \geq 1$.

(2) $\implies$ (3): Assume now that $\ell(A) \geq 1$ and $\deg A = 0$. Then $A \sim A'$ for some $A' \geq 0$ (by Remark 5.2(b)). The conditions $A' \geq 0$ and $\deg A' = 0$ imply that $A' = 0$, hence $\ell(A) = \ell(A') = \ell(0) = 1$ by Lemma 5.4(a) $\mathcal{L}(0) = K$.

(3) $\implies$ (1): Let $\ell(A) = 1$ and $\deg A = 0$. Choose $0 \neq z \in \mathcal{L}(A)$, then $(z) + A \geq 0$. As $\deg((z) + A) = 0$ as they are similar, it follows that $(z) + A = 0$, hence $A = -(z) = (z^{-1})$ is principal. $\square$

Definition 5.10. The genus g of F/K is defined by

$$g := \max \{ \deg A - \ell(A) + 1 \mid A \in \text{Div}(F) \}$$

Corollary 5.11. The genus of F/K is a non-negative integer.

Proof. In the definition of g , put $A = 0$. Then $\deg(0) - \ell(0) + 1 = 0$, hence $g \geq 0$. $\square$

Theorem 5.12 (Reimann's Theorem). Let F/K be a function field of genus g . Then we have:

(a) For all divisors $A \in \text{Div}(F)$,

$$\ell(A) \geq \deg A + 1 - g.$$

(b) There is an integer c , depending only on the function field F/K , such that

$$\ell(A) = \deg A + 1 - g$$

whenever $\deg A \leq c$.

Proof. (a) From definition of the genus.

(b) Choose a divisor A_0 with $g = \deg A_0 - (A_0) + 1$ and set $c := \deg A_0 + g$. If $\deg A \geq c$ then $(A - A_0) \geq \deg(A - A_0) + 1 - g \leq c - \deg A_0 + 1 - g = 1$. So there is an element $0 \neq z \in \mathcal{L}(A - A_0)$. Consider the divisor $A := A + (z) \leq A_0$. We have $\deg A - \ell(A) = \deg A' - \ell(A') \leq \deg A_0 - \ell(A_0) = g - 1$ by Reimann's Theorem. Hence $\ell(A) \leq \deg A + 1 - g$. $\square$

Example 5.13. *Rational function field $K(x)/K$ has genus $g = 0$.*

Proof. Let P_∞ denote the pole divisor of x . Consider for $r \geq 0$ the vector space $\mathcal{L}(rP_\infty)$. Obviously the elements $1, x, \dots, x^r$ are in $\mathcal{L}(rP_\infty)$, hence $r + 1 \leq \ell(rP_\infty) = \deg(rP_\infty) + 1 - g = r + 1 - g$ for sufficiently large r . Hence $g \leq 0$. As $g \geq 0$ holds for every function field, $g = 0$. $\square$

Chapter 6

Adele

Definition 6.1. *An adele of F/K is a mapping*

$$\alpha : \begin{cases} \mathbb{P}_F \rightarrow F \\ P \mapsto \alpha_P \end{cases}$$

such that $\alpha_P \in O_P$ for almost all $P \in \mathbb{P}_F$. By definition we have that $v_P(\alpha) \geq 0$ for almost all $P \in \mathbb{P}_F$. We regard an adele as an element of the direct product $\prod_{P \in \mathbb{P}_F} F$ and hence use the notation $\alpha = (\alpha_P)_{P \in \mathbb{P}_F}$ or, even shorter, $\alpha = (\alpha_P)$. The set

$$A_F := \{\alpha \mid \alpha \text{ is an adele of } F/K\}$$

is called the adele space of F/K . The principal adele of an element $x \in F$ is the adele all of whose components are equal to x (note that this definition makes sense since x has only finitely many poles). This gives an embedding $F \rightarrow A_F$. The valuations v_P of F/K extend naturally to A_F by setting

$$v_P(\alpha) := v_P(\alpha_P) \text{ (where } \alpha_P \text{ is the } P\text{-component of the adele } \alpha).$$

It is regarded as a vector space over K in the obvious manner (actually A_F can be regarded as a ring, but the ring structure will never be used), we define one of its K -subspace as:-

Definition 6.2. *For $A \in \text{Div}(F)$ we define*

$$A_F(A) := \{\alpha \in A_F \mid v_P(\alpha) \geq -v_P(A) \text{ for all } P \in \mathbb{P}_F\}.$$

Remark 6.3. *Note that although the vector spaces $A_F, A_F(A)$ and F are infinite-dimensional, the theorem below states that the quotient space $A_F/(A_F(A) + F)$ has finite dimension over K . As a corollary, we obtain another characterization of the genus of F/K .*

Corollary 6.4. $g = \dim(A_F/(A_F(0) + F))$

Proof. $i(0) = \ell(0) - \deg(0) + g - 1 = 1 - 0 + g - 1 = g$. □

Theorem 6.5. *For every divisor A the index of specialty is*

$$i(A) = \dim(A_F/(A_F(A) + F)).$$

Here, as usual, $\dim$ means the dimension as a K -vector space.

Remark 6.6. 1. *Theorem 6.5 can be restated as follows: for all $A \in \text{Div}(F)$ holds*

$$\ell(A) = \deg A + 1 - g + \dim(A_F/(A_F(A) + F)) \tag{6.1}$$

2. *Next we introduce Weils' Differentials, which lead to second interpretation of index of speciality of a divisor.*

Chapter 7

The Weil Differential

NOTE: F/K denotes algebraic function field of genus g .

Definition 7.1. For $A \in \text{Div}(F)$ the integer

$$i(A) := \ell(A) - \deg A + g - 1$$

is called the index of specialty of A .

Remark 7.2. By Riemann's Theorem 5.12(a) we get that $i(A) \geq 0$ and by part (b) we get $i(A) = 0$ $\deg A$ is sufficiently large.

Definition 7.3. A Weil differential of F/K is a K -linear map $\omega : A_F \rightarrow K$ vanishing on $A_F(A) + F$ for some divisor $A \in \text{Div}(F)$. We call

$$\Omega_F := \{\omega \mid \omega \text{ is a Weil differential of } F/K\}$$

the module of Weil differentials of F/K . For $A \in \text{Div}(F)$. let

$$\Omega_F(A) := \{\omega \in \Omega_F \mid \omega \text{ vanishes on } A_F(A) + F\}.$$

Remark 7.4. 1. Ω_F is a K -vector space w.r.t. function addition and multiplication.

2. if ω_1 vanishes on $A_F(A_1) + F$ and ω_2 vanishes on $A_F(A_2) + F$ then $\omega_1 + \omega_2$ vanishes on $A_F(A_3) + F$ for every divisor $A_3 \leq A_1$ and $A_3 \geq A_2$ as A_3 is contained in both, and a ω_1 vanishes on $A_F(A_1) + F$ for $a \in K$.

3. $\Omega_F(A)$ is a subspace of Ω_F .

Lemma 7.5. For $A \in \text{Div}(F)$ we have $\dim \omega_F(A) = i(A)$.

Remark 7.6. A simple consequence of this lemma is $\Omega_F \neq 0$.

Choose a divisor A of degree ≤ -2 .

$\implies \dim \Omega_F(A) = i(A) = \ell(A) - \deg A + g - 1 \geq 1$, hence $\Omega_F(A) \neq 0$.

Definition 7.7. For $x \in F$ and $\omega \in \Omega_F$ we define

$$x\omega : \begin{cases} A_F \rightarrow K \\ (x\omega)(\alpha) \mapsto \omega(x\alpha) \end{cases}$$

Remark 7.8. $x\omega$ is again a Weil differential of F/K . In fact, if ω vanishes on $A_F(A) + F$ then $x\omega$ vanishes on $A_F(A + (x)) + F$.

$A_F(A + (x)) = \{\alpha \in A_F \mid v_P(\alpha) \geq -v_P(A + (x)) \text{ for all } P \in \mathbb{P}_F\}$ Let $(x\omega)(\alpha) = \omega(x\alpha) = 0 \iff x\alpha \in A_F(A) + F$.

Proposition 7.9. Ω_F is a one-dimensional vector space over F .

Proof. Choose $0 \neq \omega_1 \in \Omega_F$ (we already know that $\Omega_F \neq 0$). We need to prove that $\forall \omega_2 \in \Omega_F \exists z \in F$ with $\omega_2 = z\omega_1$. Let $\omega_2 \neq 0$. Choose $A_1, A_2 \in \text{Div}(F)$ such that $\omega_1 \in \omega_F(A_1)$ and $\omega_2 \in \omega_F(A_2)$. For a divisor B consider the K -linear injective maps

$$p_i : \begin{cases} (A_i + B) \rightarrow \Omega_F(-B), \\ x \mapsto x\omega_i. \end{cases} \quad (i = 1, 2)$$

Claim. For an appropriate choice of the divisor B ,

$$p_1(\mathcal{L}(A_1 + B)) \cap p_2(\mathcal{L}(A_2 + B)) \neq \{0\}.$$

Proof of the Claim. Let U_1, U_2 be subspaces of V then,

$$\dim(U_1 + U_2) \leq \dim(V) \implies \dim(U_1 \cap U_2) \geq \dim(U_1) + \dim(U_2) - \dim(V). \quad (7.1)$$

Let $B > 0$ be a divisor such that by Reimann's Theorem:-

$$\ell(A_i + B) = \deg(A_i + B) + 1 - g$$

for $i = 1, 2$. Let $U_i := p_i(\mathcal{L}(A_i + B)) \subseteq \Omega_F(-B)$. As $\dim \Omega_F(-B) = i(-B) = \ell(-B) - \deg(-B) + g - 1 = \deg B - 1 + g$ [$\deg(B) < 0 \implies \ell(B) = 0$ by Corollary 5.9(b)].

Hence $\dim U_1 + \dim U_2 - \dim \Omega_F(B) = \deg(A_1 + B) + 1 - g + \deg(A_2 + B) + 1 - g - (\deg B + g - 1) = \deg B = \deg B + (\deg A_1 + \deg A_2 + 3(1 - g)) > 0$. If $\deg B$ is sufficiently large. By (7.1) it follows that $U_1 \cap U_2 \neq \{0\}$ which proves our claim.

Using this claim, we choose $x_1 \in \mathcal{L}(A_1 + B)$ and $x_2 \in \mathcal{L}(A_2 + B)$ such that $x_1\omega_1 = x_2\omega_2 \neq 0$. Then $\omega_2 = (x_1x_2^{-1})\omega_1$ which proves the Proposition. $\square$

NOTE: We want to attach a divisor to each Weil differential $\omega \neq 0$. To this end we consider (for a fixed ω) the set of divisors

$$M(\omega) := \{A \in \text{Div}(F) \mid \omega \text{ vanishes on } A_F(A) + F\}.$$

Lemma 7.10. *Let $0 \neq \omega \in \Omega_F$. Then there is a uniquely determined divisor $W \in M(\omega)$ such that $A \leq W$ for all $A \in M(\omega)$.*

Theorem 7.11 (Duality Theroem). *Let A be an arbitrary divisor and $W = (\omega)$ be a canonical divisor of F/K . Then the mapping*

$$\mu : \begin{cases} (W - A) \rightarrow \omega_F(A), \\ x \mapsto x\omega \end{cases}$$

is an isomorphism of K -vector spaces. In particular,

$$i(A) = \ell(W - A).$$

Proof. For $x \in \mathcal{L}(W - A)$ we have $(x\omega) = (x) + (\omega) \geq -(W - A) + W = A$, hence $x\omega \in \omega_F(A)$. Hence μ maps $\mathcal{L}(W - A)$ into $\omega_F(A)$ and is linear and injective. In order to show that μ is onto, we consider a Weil differential $\omega_1 \in \Omega_F(A)$.

$\omega_1 = x\omega$ with some $x \in F$. As $(x) + W = (x) + (\omega) = (x\omega) = (\omega_1) \geq A \implies (x) \geq -(W - A)$, so $x \in \mathcal{L}(W - A)$ and $\omega_1 = \mu(x)$. Hence $\dim \omega_F(A) = \ell(W - A) = i(A)$. $\square$

Chapter 8

Riemann-Roch Theorem

Theorem 8.1 (Riemann-Roch Theorem). *Let W be a canonical divisor of F/K . Then for each divisor $A \in \text{Div}(F)$,*

$$\ell(A) = \deg A + 1 - g + \ell(W - A).$$

Proof. Obvious by Theorem 7.11 $i(A) = \ell(W - A) = \ell(A) - \deg A + g - 1$. $\square$

Corollary 8.2. *For a canonical divisor W we have $\deg W = 2g - 2$ and $\ell(W) = g$.*

Proof. For $A = 0$, the Riemann-Roch Theorem gives $1 = \ell(0) = \deg 0 + 1 - g + \ell(W - 0) \implies \ell(W) = g$. Setting $A = W$ we obtain $g = \ell(W) = \deg W + 1 - g + \ell(W - W) = \deg W + 2 - g$. Therefore $\deg W = 2g - 2$. $\square$

Theorem 8.3. *If A is a divisor of F/K of degree $\deg A \geq 2g - 1$ then $\ell(A) = \deg A + 1 - g$.*

Proof. We have $\ell(A) = \deg A + 1 - g + \ell(W - A)$, where W is a canonical divisor. As $\deg A \geq 2g - 1$ and $\deg W = 2g - 2$, we conclude that $\deg(W - A) < 0$. Lemma 5.9(b) $\implies \ell(W - A) = 0$. $\square$

Remark 8.4. *Observe that the bound $2g - 1$ in this theorem is the best possible, since for a canonical divisor W , $\ell(W) > \deg W + 1 - g$.*

NOTE: F/K denotes an algebraic function field of genus g . Some useful characterisations of canonical divisors:-

Proposition 8.5. *Let $g_0 \in \mathbb{Z}$ and $W_0 \in \text{Div}(F)$ satisfy*

$$\ell(A) = \deg A + 1 - g_0 + (W_0 - A) \quad (8.1)$$

for all $A \in \text{Div}(F)$. Then $g_0 = g$, and W_0 is a canonical divisor.

Proof. Setting $A = 0$ resp. $A = W_0$ in (8.1) we obtain $\ell(W_0) = g_0$ and $\deg W_0 = 2g_0 - 2, \dots (1)$ (by Corollary 8.2).

Let W be a canonical divisor of F/K . We choose a divisor A with $\deg A > \max\{2g - 2, 2g_0 - 2\}$. Then $\ell(A) = \deg A + 1 - g$ by Theorem 6.4 and hence by (8.1) $g = g_0$.

We substitute (1) and $A = W$ in (8.1). This gives $g = (2g - 2) + 1 - g + (W_0 - W)$ hence $\ell(W_0 - W) = 1$. As $\deg(W_0 - W) = 0$, this implies that $W_0 - W$ is principal. So $W_0 - W = (x) \implies W_0 \implies W$. $\square$

Proposition 8.6. *A divisor C is canonical $\iff \deg C = 2g - 2$ and $\ell(C) \geq g$.*

Proof. Let $\deg C = 2g - 2$ and $\ell(C) \geq g$. Choose a canonical divisor W . Then $g \leq \ell(C) = \deg C + 1 - g + \ell(W - C) = g - 1 + \ell(W - C)$ hence $\ell(W - C) \geq 1$. As $\deg(W - C) = 0$, it follows from Corollary 5.2 that $W \sim C$.

Conversely, let C be canonical. Result holds by Corollary 8.2. $\square$

Proposition 8.7. *For a function field F/K the following conditions are equivalent:*

(1) *F/K is rational; i.e., $F = K(x)$ for some x which is transcendental over the field K .*

(2) *F/K has genus 0, and there is some divisor $A \in \text{Div}(F)$ with $\deg A = 1$.*

Proof. (1) $\implies$ (2): $g = 0$ for P_α which is a divisor of degree 1.

(2) $\implies$ (1): Let $g = 0$ and $\deg A = 1$. As $\deg A \geq 2g - 1$ we have that $\ell(A) = \deg A + 1 - g = 2$. Hence $A \sim A'$ for some positive divisor A' (see Remark 5.2(b)). As $\ell(A') = 2 \exists x \in \mathcal{L}(A') \setminus K$ so $(x) \neq 0$ and $(x) + A \geq 0$. As $A' \geq 0 \implies v_P(A') \geq 0 \forall$ places P and $\deg A' = 1$, this is possible only if $A' = (x)_\infty$, the pole divisor of x . Now $[F : K(x)] = \deg(x)_\infty = \deg A = 1$ by Theorem 5.7, so $F = K(x)$. $\square$

Remark 8.8. 1. *There exist non-rational function fields of genus 0 (these cannot have a divisor of degree 1 by Proposition 6.3 of [1]).*

2. *However, if K is an algebraically closed field or a finite field, there exists always a divisor of degree 1 (for an algebraically closed field this is trivial) hence in these cases we have $g = 0$ if and only if F/K is rational.*

Theorem 8.9 (Strong Approximation Theorem). *Let $S \subsetneq \mathbb{P}_F$ be a proper subset of $\mathbb{P}_F$ and $P_1, \dots, P_r \in S$. Suppose there are given elements $x_1, \dots, x_r \in F$ and integers $n_1, \dots, n_r \in \mathbb{Z}$. Then there exists an element $x \in F$ such that*

$$v_{P_i}(x - x_i) = n_i (i = 1, \dots, r), \text{ and}$$

$$v_P(x) \geq 0 \text{ for all } P \in S \setminus \{P_1, \dots, P_r\}.$$

Proposition 8.10. *Let $P \in \mathbb{P}_F$. Then for each $n \geq 2g$ there exists an element $x \in F$ with pole divisor $(x)_\infty = nP$.*

Proof. $-nP \leq (x) < -(n-1)P \implies (x) = -nP$. □

Definition 8.11. *Let $P \in \mathbb{P}_F$. An integer $n \geq 0$ is called a pole number of P if there is an element $x \in F$ with $(x)_\infty = nP$. Otherwise n is called a gap number of P .*

Remark 8.12. *Clearly n is a pole number of P if and only if $\ell(nP) > \ell((n-1)P)$. Moreover, the set of pole numbers of P is a sub-semigroup of the additive semigroup $\mathbb{N}$ (to see this note that, if $(x_1)_\infty = n_1P$ and $(x_2)_\infty = n_2P$ then x_1x_2 has the pole divisor $(x_1x_2)_\infty = (n_1 + n_2)P$).*

Chapter 9

Local Components of Weil Differentials

Previously we considered the diagonal embedding $F \rightarrow A_F$ which maps $x \in F$ to the corresponding principal adele. Now we introduce for each place $P \in \mathbb{P}_F$ another local embedding $\iota_P : F \rightarrow A_F$.

Definition 9.1. *Let $P \in \mathbb{P}_F$.*

- (a) *For $x \in F$ let $\iota_P(x) \in A_F$ be the adele whose P -component is x , and all other components are 0.*
- (b) *For a Weil differential $\omega \in \Omega_F$ we define its local component $\omega_P : F \rightarrow K$ by $\omega_P(x) := \omega(\iota_P(x))$. Clearly ω_P is a K -linear mapping.*

Proposition 9.2. *Let $\omega \in \omega_F$ and $\alpha = (\alpha_P) \in A_F$. Then $\omega_P(\alpha_P) = 0$ for at most finitely many places P , and*

$$\omega(\alpha) = \sum_{P \in \mathbb{P}_F} \omega_P(\alpha_P).$$

In particular

$$P \in \mathbb{P}_F \omega_P(1) = 0. \tag{9.1}$$

Proof. Let $\omega = 0$ and $W = (\omega)$, the divisor of ω . There is a finite set $S \subseteq \mathbb{P}_F$ such that

$$v_P(W) = 0 \text{ and } v_P(\alpha_P) \geq 0 \text{ for all } P \in S .$$

Define $b = (b_P) \in A_F$ by

$$b_P := \begin{cases} P & \text{for } P \notin S, \\ 0 & \text{for } P \in S. \end{cases}$$

Then $b \in A_F(W)$ and $\alpha = b + \sum_{P \in S} \iota_P(\alpha_P) \implies \omega(b) = 0$ and $\omega(\alpha) = \sum_{P \in S} \omega_P(\alpha_P)$. For $P \in S$, $\iota_P(\alpha_P) \in A_F(W)$ and hence $\omega_P(\alpha_P) = 0$. $\square$

Proposition 9.3. (a) Let $\omega \neq 0$ be a Weil differential of F/K and $P \in \mathbb{P}_F$. Then $v_P(\omega) = \max \{r \in \mathbb{Z} \mid \omega_P(x) = 0 \forall x \in F \text{ with } v_P(x) \geq -r\}$. In particular ω_P is not identically 0.

(b) If $\omega, \omega' \in \Omega_F$ and $\omega_P = \omega'_P$ for some $P \in \mathbb{P}_F$ then $\omega = \omega'$.

Proof. (a) By definition, $v_P(\omega) = v_P(W)$ where $W = (\omega)$ denotes the divisor of ω . Let $s := v_P(\omega)$. For $x \in F$ with $v_P(x) \geq -s$ we have $\iota_P(x) \in A_F(W)$, hence $\omega_P(x) = \omega(\iota_P(x)) = 0$. Let $\omega_P(x) = 0$ for all $x \in F$ with $v_P(x) \geq -s - 1$. Let $\alpha = (\alpha_Q)_{Q \in \mathbb{P}_F} \in A_F(W + P)$. Then $\alpha = (\alpha - \iota_P(\alpha_P)) + \iota_P(\alpha_P)$ with $\alpha - \iota_P(\alpha_P) \in A_F(W)$ and $v_P(\alpha_P) \geq -s - 1$, hence $\omega(\alpha) = \omega(\alpha - \iota_P(\alpha_P)) + \omega_P(\alpha_P) = 0$. Therefore ω vanishes on $A_F(W + P)$, a contradiction to the definition of W .

(b) If $\omega_P = \omega'_P$ then $(\omega - \omega')_P = 0$, hence $\omega - \omega' = 0$ by (a). $\square$

Proposition 9.4. For the rational function field $F = K(x)$ the following hold:

(a) The divisor $-2P_\infty$ is canonical.

(b) There exists a unique Weil differential $\eta \in \Omega_{K(x)}$ with $(\eta) = -2P_\infty$ and $\eta P_\infty(x^{-1}) = -1$.

(c) The local components ηP_∞ resp. ηP_α of the above Weil differential η satisfy

$$\eta P_\infty((x - a)^n) = \begin{cases} 0 & \text{for } n \neq -1, \\ -1 & \text{for } n = -1 \end{cases}$$

$$\eta P_\alpha((x - a)^n) = \begin{cases} 0 & \text{for } n \neq -1, \\ 1 & \text{for } n = -1 \end{cases}$$

Proof. (a) $\deg(-2P_\infty) = -2 = 2g - 2$ and $\ell(-2P_\infty) = 0 = g$, hence $-2P_\infty$ is canonical by Proposition 8.6.

(b) Choose a Weil differential ω with divisor $(\omega) = -2P_\infty$. Then ω vanishes on the space $\mathcal{A}_{K(x)}(-2P_\infty)$, but it does not vanish identically on $\mathcal{A}_{K(x)}(-P_\infty)$. Since

$$\dim \mathcal{A}_{K(x)}(-P_\infty) / \mathcal{A}_{K(x)}(-2P_\infty) = 1$$

(see Equation (1.24) in the proof of Theorem 1.5.4 in [1]) and

$$\begin{aligned} \iota_{P_\infty}(x^{-1}) &\in \mathcal{A}_{K(x)}(-P_\infty) \setminus \mathcal{A}_{K(x)}(-2P_\infty), \\ \implies \omega_{P_\infty}(x^{-1}) &= \omega(\iota_{P_\infty}(x^{-1})) =: c \neq 0. \end{aligned}$$

Setting $\eta := -c^{-1}\omega$ we obtain $(\eta) = -2P_\infty$ and $\eta_{P_\infty}(x^{-1}) = -1$.// Uniqueness of η : If η^* has the same properties as η then $\eta - \eta^*$ vanishes on the space $\mathcal{A}_{K(x)}(-P_\infty) \implies \eta - \eta^* = 0$.

(c) Since a Weil differential vanishes on principal adeles, by Proposition 9.2

$$0 = \eta((x-a)^n) = \sum_{P \in \mathbb{P}_{K' (=)}} \eta_P((x-a)^n). \quad (9.2)$$

For $P \neq P_\infty$ and $P \neq P_a$ holds $v_P((x-a)^n) = 0$, therefore $\eta_P((x-a)^n) = 0$ by Proposition 9.3, and then (9.2) yields

$$\eta_{P_\infty}((x-a)^n) + \eta_{P_a}((x-a)^n) = 0. \quad (9.3)$$

In the case $n \leq -2$ we have $v_{P_\infty}((x-a)^n) \geq 2$, hence $\eta_{P_\infty}((x-a)^n) = 0$ by Proposition 9.3. Now (9.3) $\implies \eta_{P_n}((x-a)^n) = 0$ as well. If $n \geq 0$, $\eta_{P_a}((x-a)^n) = 0$ by Proposition 9.3, and we obtain the result for $\eta_{P_\infty}((x-a)^n)$ again by (9.3). Eventually we consider the case $n = -1$. Since

$$\frac{1}{x-a} = \frac{a}{x(x-a)} + \frac{1}{x} \quad \text{and} \quad \iota_{P_\infty}\left(\frac{a}{x(x-a)}\right) \in \mathcal{A}_{K(x)}(-2P_\infty),$$

we see that $\eta_{P_\infty}((x-a)^{-1}) = \eta_{P_\infty}(x^{-1}) = -1$ (by definition of η), and from (9.3) follows that $\eta_{P_a}((x-a)^{-1}) = 1$. $\square$

Chapter 10

Introduction To Codes

We now study the basic theory of algebraic codes, required for studying Algebraic Geometry Codes (AG) Later.

Let $\mathbb{F}_q$ denote the finite field with q elements. We consider the n dimensional vector space $\mathbb{F}_q^n$ whose elements are n -tuples $a = (a_1, \dots, a_n)$ with $a_i \in \mathbb{F}_q$.

Definition 10.1. Let $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n) \in \mathbb{F}_q^n$. The function defined as

$$d(a, b) := |\{i; a_i \neq b_i\}|$$

is called the Hamming distance on $\mathbb{F}_q^n$.

Definition 10.2. The weight of an element $a \in \mathbb{F}_q^n$ is defined as

$$\text{wt}(a) := d(a, 0) = |\{i; a_i \neq 0\}|.$$

Remark 10.3. The Hamming distance is a metric on $\mathbb{F}_q^n$. We give proof for Triangle Inequality:

$d(a, c) \leq d(a, b) + d(b, c) \forall a, b, c \in \mathbb{F}_q^n$. Let (a, b) be common in l_1 places and (b, c) in l_2 places.

(a, c) are common at maximum $l_3 = \max l_1, l_2$. Clearly $n - l_3 < n - l_2 \leq n - l_2 + n - l_3 \implies d(a, c) \leq d(a, b) + d(b, c)$. Other properties are trivial.

Definition 10.4. A code C is a linear subspace of $\mathbb{F}_q^n$. The elements of C are called codewords. Following terms define a code:

1. We call n the length of C and $\dim C$ (as a $\mathbb{F}_q$ -vector space) the dimension of C .
2. An $[n, k]$ code is a code of length n and dimension k . The minimum distance $d(C)$ of a code $C \neq 0$ is defined as

$$d(C) := \min\{d(a, b) \mid a, b \in C \text{ and } a \neq b\} = \min\{wt(c) \mid 0 \neq c \in C\}.$$

3. An $[n, k]$ code with minimum distance d will be referred to as an $[n, k, d]$ code.

Remark 10.5. 1. If $C \subseteq \mathbb{F}_q^n$ is a linear subspace, C is said to be a linear code. Most codes used in practice belong to this class, hence we will only consider linear codes.

2. For a code C with minimum distance $d = d(C)$ we set $t := \lfloor (d-1)/2 \rfloor$ (where $\lfloor x \rfloor$ denotes the integer part of the real number x). Then C is said to be t -error correcting. Clearly if $u \in \mathbb{F}_q^n$ and $d(u, c) \leq t$ for some $c \in C$ then c is the only code-word with $d(u, c) \leq t$.
3. A simple way to describe a specific code C explicitly is to write down a basis of C (as a vector space over $\mathbb{F}_q$).

Definition 10.6. Let C be an $[n, k]$ code over $\mathbb{F}_q$. A generator matrix of C is a $k \times n$ matrix whose rows are a basis of C .

Definition 10.7. The canonical inner product on $\mathbb{F}_q^n$ is defined by

$$\langle a, b \rangle := \sum_{i=1}^n a_i b_i,$$

for $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n) \in \mathbb{F}_q^n$.

Definition 10.8. If $C \subseteq \mathbb{F}_q^n$ is a code then

$$C^\perp := \{u \in \mathbb{F}_q^n \mid \langle u, c \rangle = 0 \text{ for all } c \in C\}$$

is called the dual of C . The code C is self-dual if $C = C^\perp$ (resp. $C \subseteq C^\perp$).

As $C \odot C^\perp = F_q^n$, dual of an $[n, k]$ code is an $[n, n-k]$ code, and $(C^\perp)^\perp = C$. In particular, the dimension of a self-dual code of length n is $n/2$.

Definition 10.9. A generator matrix of C is a matrix whose rows form a basis for C .

Definition 10.10. A generator matrix H of $C^\perp$ is said to be a parity check matrix for C .

Clearly a parity check matrix H of an $[n, k]$ code C is an $(n-k) \times n$ matrix of rank $n-k$, and we have

$$C = \{u \in \mathbb{F}_q^n \mid H \cdot u^t = 0\}$$

(where u^t denotes the transpose of u). Hence a parity check matrix 'checks' whether a vector $u \in \mathbb{F}_q^n$ is a codeword or not.

Definition 10.11. Two codes $C_1, C_2 \subseteq \mathbb{F}_q^n$ are said to be equivalent if there is a vector $a = (a_1, \dots, a_n) \in (\mathbb{F}_q^\times)^n$ such that $C_2 = a \cdot C_1$; i.e.,

$$C_2 = \{(a_1 c_1, \dots, a_n c_n) \mid (c_1, \dots, c_n) \in C_1\}.$$

Evidently equivalent codes have the same dimension and the same minimum distance. One of the basic problems in algebraic coding theory is to constructing codes whose dimension and minimum distance are large in comparison with their length given a fixed alphabet $\mathbb{F}_q$. If the dimension of a code is large (with respect to its length), then its minimum distance is small. The simplest bound is the Singleton Bound, explained below.

Proposition 10.12 (Singleton Bound). For an $[n, k, d]$ code C holds

$$k + d \leq n + 1.$$

Proof. Consider the linear subspace

$$E := \{(a_1, \dots, a_n) \in \mathbb{F}_q^n \mid a_i = 0 \text{ for all } i \geq d\} \subseteq \mathbb{F}_q^n.$$

$\forall a \in E \text{ wt}(a) \leq d-1 \implies E \cap C = \{0\}$. As $\dim E = d-1$,

$$\begin{aligned} \implies k + (d-1) &= \dim C + \dim E \\ &= \dim(C + E) + \dim(C \cap E) = \dim(C + E) \leq n. \end{aligned}$$

□

Codes with $k + d = n + 1$ are called MDS codes (maximum distance separable codes). It is difficult to obtain lower bounds for the minimum distance of a given code. Algebraic geometry codes (AG codes) were introduced by V.D. Goppa in [3]. For this class of codes a good lower bound for the minimum distance is available. Algebraic geometry codes are a very natural generalization of Reed-Solomon, discussed below.

Reed Solomon Codes

Let $n = q - 1$. Let $b \in \mathbb{F}_q$ be a primitive element of the multiplicative group $\mathbb{F}_q^\times$; i.e., $\mathbb{F}_q^\times = \{b^1, b^2, \dots, b^n = 1\}$. For $k \in \mathbb{Z}$ with $1 \leq k \leq n$ we consider the k -dimensional vector space

$$\mathcal{L}_k := \{f \in \mathbb{F}_q[X] \mid \deg f \leq k - 1\}$$

and the evaluation map $\text{ev} : \mathcal{L}_k \rightarrow \mathbf{F}_q^n$ given by

$$\text{ev}(f) := (f(b), f(b^2), \dots, f(b^n)) \in \mathbb{F}_q^n. \quad (10.1)$$

This map is

1. $\mathbb{F}_q$ -linear.
2. Injective. Finding Kernel of the map: $\text{ev}(f) := (f(b), f(b^2), \dots, f(b^n)) = 0 \iff f(b^i) = 0 \forall 1 \leq i \leq n$. As a non-zero polynomial $f \in \mathbb{F}_q[X]$ of degree $< n$ has less than n zeros. Hence f is identically zero.

$\implies C_k := \{(f(b), f(b^2), \dots, f(b^n)) \mid f \in \mathcal{L}_k\}$ is an $[n, k]$ code over $\mathbb{F}_q$; it is called an RS code (Reed-Solomon code).

The weight of a codeword $0 \neq c = \text{ev}(f) \in C_k$ is given by

$$\begin{aligned} \text{wt}(c) &= n - |\{i \in \{1, \dots, n\}; f(b^i) = 0\}| \\ &\geq n - \deg f \geq n - (k - 1). \end{aligned}$$

Hence the minimum distance d of C_k satisfies the inequality $d \geq n + 1 - k$. Also $d \leq n + 1 - k$ by the Singleton Bound. Hence RS codes are MDS codes over $\mathbb{F}_q$. Observe however that RS codes are short in comparison with the size of the alphabet $\mathbb{F}_q$, as $n = q - 1$.

Chapter 11

Algebraic Geometry Codes (AG Codes)

Now we introduce algebraic geometry codes. Notations:

1. $F/\mathbb{F}_q$: an algebraic function field of genus g .
2. $P_1, \dots, P_n$: pairwise distinct places of $F/\mathbb{F}_q$ of degree 1.
3. $D = P_1 + \dots + P_n$.
4. G : is a divisor of $F/\mathbb{F}_q$ such that $\text{supp } G \cap \text{supp } D = \emptyset$.

Definition 11.1. *The algebraic geometry code (or AG code) $C_{\mathcal{L}}(D, G)$ associated with the divisors D and G is defined as*

$$C_{\mathcal{L}}(D, G) := \{(x(P_1), \dots, x(P_n)) \mid x \in \mathcal{L}(G)\} \subseteq \mathbb{F}_q^n.$$

Remark 11.2. *This code is defined as $x \in \mathcal{L}(G) \implies (x) = (x)_0 - (x)_\infty \geq -G \implies v_{P_i}(x) \geq 0$ ($i = 1, \dots, n$) as $\text{supp } G \cap \text{supp } D = \emptyset \implies v_{P_i}(G) \geq 0 \forall i$. The residue class $x(P_i)$ of x modulo P_i is an element of the residue class field of P_i . As $\deg P_i = 1$, this residue class field is $\mathbb{F}_q$, so $x(P_i) \in \mathbb{F}_q$.*

As in (10.1) we can consider the evaluation map, analogous with the definition of Reed-Solomon codes: $\text{ev } D : \mathcal{L}(G) \rightarrow \mathbb{F}_q^n$ given by

$$\text{ev } D(x) := (x(P_1), \dots, x(P_n)) \in \mathbb{F}_q^n. \quad (11.1)$$

Theorem 11.3. $C_{\mathcal{L}}(D, G)$ is an $[n, k, d]$ code with parameters

$$k = \ell(G) - \ell(G - D) \quad \text{and} \quad d \geq n - \deg G.$$

Proof. The evaluation map (11.1) is onto linear map from $\mathcal{L}(G)$ to $C_{\mathcal{L}}(D, G)$ with

$$\text{Ker}(\text{ev } D) = \{x \in \mathcal{L}(G) \mid x(P_i) = 0 \implies v_{P_i}(x) > 0 \text{ for } i = 1, \dots, n\} = \mathcal{L}(G - D).$$

$\implies k = \dim C_{\mathcal{L}}(D, G) = \dim \mathcal{L}(G) - \dim \mathcal{L}(G - D) = \ell(G) - \ell(G - D)$. Let $C_{\mathcal{L}}(D, G) \neq 0$ so that minimum distance d makes senses. Choose an element $x \in \mathcal{L}(G)$ with $\text{wt}(\text{ev}_D(x)) = d$. Then exactly $n - d$ places $P_{i_1}, \dots, P_{i_{n-d}}$ in the support of D are zeros of x , so

$$0 \neq x \in \mathcal{L}(G - (P_{i_1} + \dots + P_{i_{n-d}})).$$

Corollary 5.9 implies

$$0 \leq \deg(G - (P_{i_1} + \dots + P_{i_{n-d}})) = \deg G - n + d.$$

$$\implies d \geq n - \deg G. \quad \square$$

Corollary 11.4. *Let the degree of $G < n$. Then the evaluation map $\text{ev } D : \mathcal{L}(G) \rightarrow C_{\mathcal{L}}(D, G)$ is injective, and we have:*

Proof. Let $d^* = d \implies \exists 0 \neq x \in \mathcal{L}(G)$ such that the codeword $(x(P_1), \dots, x(P_n)) \in C_{\mathcal{L}}(D, G)$ has precisely $n - d = n - d^* = \deg G$ zero components, say $x(P_{i_j}) = 0$ for $j = 1, \dots, \deg G$. Put

$$D' := \sum_{j=1}^{\deg G} P_{i_j}.$$

Then $0 \leq D' \leq D$, $\deg D' = \deg G$ and $\ell(G - D') > 0$ (as $x \in \mathcal{L}(G - D')$). Conversely, if D' has the above properties then we choose an element $0 \neq y \in \mathcal{L}(G - D')$. The weight of the corresponding codeword $(y(P_1), \dots, y(P_n))$ is $n - \deg G = d^*$, hence $d = d^*$. $\square$

Chapter 12

AG Codes Using Weil Differentials

Another code can be associated with the divisors G and D , by using local components of Weil differentials as in Section 9. These are called Algebraic Geometry Codes.

Definition 12.1. *Let G and $D = P_1 + \dots + P_n$ be divisors and $\text{supp } D = \emptyset$. Then define the code $C_\Omega(D, G) \subseteq \mathbb{F}_q^n$ by*

$$C_\Omega(D, G) := \{(\omega_{P_1}(1), \dots, \omega_{P_n}(1)) \mid \omega \in \Omega_F(G - D)\}.$$

The code $C_\Omega(D, G)$ is called an algebraic geometry code.

Theorem 12.2. *1. $C_\Omega(D, G)$ is an $[n, k', d']$ code with parameters*

$$k' = i(G - D) - i(G) \quad \text{and} \quad d' \geq \deg G - (2g - 2).$$

If $\deg G > 2g - 2$, we have $k' = i(G - D) \geq n + g - 1 - \deg G$.

2. $2g - 2 < \deg G < n \implies k' = n + g - 1 - \deg G$.

Proof. 1. Let $P \in \mathbb{P}_F$ be a place of degree one and let ω be a Weil differential with $v_P(\omega) \geq -1$.

$$\text{Claim: } \omega_P(1) = 0 \iff v_P(\omega) \geq 0. \quad (12.1)$$

Proof Of Claim. ($\Leftarrow$) proof trivial as $v_P(1) = 0$ and by Proposition 9.3, which tells for an integer $r \in \mathbb{Z}$.

$$v_P(\omega) \geq r \iff \omega_P(x) = 0 \text{ for all } x \in F \text{ with } v_P(x) \geq -r. \quad (12.2)$$

($\Rightarrow$) Let $\omega_P(1) = 0$. Let $x \in F$ with $v_P(x) \geq 0$. As $\deg P = 1$, we can write $x = a + y$ with $a \in \mathbb{F}_q$ and $v_P(y) \geq 1$.

$$\implies \omega_P(x) = \omega_P(a) + \omega_P(y) = a \cdot \omega_P(1) + 0 = 0. \quad (12.3)$$

Observe that $\omega_P(y) = 0$ because $v_P(\omega) \geq -1$ and $v_P(y) \geq 1$, cf. (12.1). Hence (12.2) is proved.

Now consider the $\mathbb{F}_q$ -linear mapping

$$\rho_D : \begin{cases} \Omega_F(G - D) & \longrightarrow C_\Omega(D, G), \\ \omega & \longmapsto (\omega_{P_1}(1), \dots, \omega_{P_n}(1)). \end{cases}$$

ρ_D is onto, and $(\text{Ker}(\rho_D)) = \Omega_F(G)$.

$$\implies k' = \dim \Omega_F(G - D) - \dim \Omega_F(G) = i(G - D) - i(G). \quad (12.4)$$

Let $\rho_D(\omega) \in C_\Omega(D, G)$ be a codeword of weight $m > 0$ and minimum distance d' . Then $\omega_{P_i}(1) = 0$ for certain $i = i_1, \dots, i_{n-m}$, so

$$\omega \in \Omega_F \left(G - \left(D - \sum_{j=1}^{n-m} P_{i_j} \right) \right)$$

by (12.1). As $\Omega_F(A) \neq 0 \implies \deg A \leq 2g - 2$ (by Theorem 8.3), we obtain

$$\begin{aligned} 2g - 2 &\geq \deg G - (n - (n - m)) = \deg G - m \\ \implies d' &\geq m \geq \deg G - (2g - 2). \end{aligned}$$

2. Now let $\deg G > 2g - 2$. Theorem 8.3 gives $i(G) = 0$. Now (12.4) and the Riemann-Roch Theorem (Theorem 8.1) give

$$\begin{aligned} k' &= i(G - D) = \ell(G - D) - \deg(G - D) - 1 + g \\ &= \ell(G - D) + n + g - 1 - \deg G. \end{aligned}$$

□

Definition 12.3. In analogy to Definition 10.1 of hamming distance the integer $\deg G - (2g - 2)$ is called the designed distance of $C_\Omega(D, G)$.

Theorem 12.4. The codes $C_{\mathcal{L}}(D, G)$ and $C_\Omega(D, G)$ are dual to each other; i.e.,

$$C_\Omega(D, G) = C_{\mathcal{L}}(D, G)^\perp.$$

Proof. Consider a place $P \in \mathbb{P}_F$ of degree one, a Weil differential ω with $v_P(\omega) \geq -1$ and an element $x \in F$ with $v_P(x) \geq 0$. Then

$$\omega_P(x) = x(P) \cdot \omega_P(1). \quad (12.5)$$

In order to prove (12.5) let $x = a + y$ with $a = x(P) \in \mathbb{F}_q$ and $v_P(y) > 0$. Then $\omega_P(x) = \omega_P(a) + \omega_P(y) = a \cdot \omega_P(1) + 0 = x(P) \cdot \omega_P(1)$.

Next we show that $C_\Omega(D, G) \subseteq C_{\mathcal{L}}(D, G)^\perp$. So let $\omega \in \Omega_F(G - D)$ and $x \in \mathcal{L}(G)$. We obtain

$$0 = \omega(x) = \sum_{P \in \mathbb{P}_F} \omega_P(x) = \sum_{i=1}^n \omega_{P_i}(x) \quad (12.6)$$

$$= \sum_{i=1}^n x(P_i) \cdot \omega_{P_i}(1) \quad (12.7)$$

$$= \langle (\omega_{P_1}(1), \dots, \omega_{P_n}(1)), (x(P_1), \dots, x(P_n)) \rangle, \quad (12.8)$$

Here $\langle, \rangle$ denotes the canonical inner product on $\mathbb{F}_q^n$. Justification for equations:

(12.6): follows from Proposition 9.2 and the fact that Weil differentials vanish on principal adeles: for $P \in \mathbb{P}_F \setminus \{P_1, \dots, P_n\}$ we have $v_P(x) \geq -v_P(\omega)$ (as $x \in \mathcal{L}(G)$ and $\omega \in \Omega(G - D)$), so $\omega_P(x) = 0$.

(12.7): follows from (12.5).

Hence $C_\Omega(D, G) \subseteq C_{\mathcal{L}}(D, G)^\perp$.

We now prove that the codes $C_\Omega(D, G)$ and $C_{\mathcal{L}}(D, G)^\perp$ have the same dimen-

sion. Using Theorems 11.3 and the Riemann-Roch Theorem 8.1:

$$\begin{aligned}
\dim C_\Omega(D, G) &= i(G - D) - i(G) \\
&= \ell(G - D) - \deg(G - D) - 1 + g - (\ell(G) - \deg G - 1 + g) \\
&= \deg D + \ell(G - D) - \ell(G) \\
&= n - (\ell(G) - \ell(G - D)) \\
&= n - \dim C_{\mathcal{L}}(D, G) = \dim C_{\mathcal{L}}(D, G)^\perp.
\end{aligned}$$

□

Our next aim is to prove that $C_\Omega(D, G)$ can be represented as $C_{\mathcal{L}}(D, H)$ with an appropriate divisor H . For this purpose we need the following lemma.

Lemma 12.5. *There exists a Weil differential η such that*

$$v_{P_i}(\eta) = -1 \text{ and } \eta_{P_i}(1) = 1 \text{ for } i = 1, \dots, n.$$

Proof. Choose an arbitrary Weil differential $\omega_0 \neq 0$. By the Weak Approximation Theorem 2.22 $\exists z \in F$ with $v_{P_i}(z) = -v_{P_i}(\omega_0) - 1$ for $i = 1, \dots, n$.

Let $\omega := z\omega_0$. We get $v_{P_i}(\omega) = -1 \implies a_i := \omega_{P_i}(1) \neq 0$ by (12.1).

Again by the Weak Approximation Theorem 2.22 we find $y \in F$ such that $v_{P_i}(y - a_i) > 0$.

$$\implies v_{P_i}(y) = 0 \text{ and } y(P_i) = a_i.$$

Put $\eta := y^{-1}\omega$ and obtain $v_{P_i}(\eta) = v_{P_i}(\omega) = -1$, and

$$\eta_{P_i}(1) = \omega_{P_i}(y^{-1}) = y^{-1}(P_i) \cdot \omega_{P_i}(1) = a_i^{-1} \cdot a_i = 1.$$

□

Proposition 12.6. *Let η be a Weil differential such that $v_{P_i}(\eta) = -1$ and $\eta_{P_i}(1) = 1$ for $i = 1, \dots, n$. Then*

$$C_{\mathcal{L}}(D, G)^\perp = C_\Omega(D, G) = C_{\mathcal{L}}(D, H) \quad \text{with} \quad H := D - G + (\eta).$$

Proof. $C_{\mathcal{L}}(D, G)^\perp = C_\Omega(D, G)$ by Theorem 12.4.

$\text{supp}(D - G + (\eta)) \cap \text{supp} D = \emptyset$ as $v_{P_i}(\eta) = -1$ for $i = 1, \dots, n$. Hence the code $C_{\mathcal{L}}(D, D - G + (\eta))$ is defined.

There is an isomorphism $\mu : \mathcal{L}(D - G + (\eta)) \rightarrow \Omega_F(G - D)$ given by $\mu(x) := x\eta$.

For $x \in \mathcal{L}(D - G + (\eta))$, $(x\eta)_{P_i}(1) = \eta_{P_i}(x) = x(P_i) \cdot \eta_{P_i}(1) = x(P_i)$

$$\implies C_\Omega(D, G) = C_{\mathcal{L}}(D, D - G + (\eta)).$$

□

Corollary 12.7. 1. Suppose there is a Weil differential η such that

$$2G - D \leq (\eta) \text{ and } \eta_{P_i}(1) = 1 \text{ for } i = 1, \dots, n.$$

Then the code $C_{\mathcal{L}}(D, G)$ is self-orthogonal; i.e., $C_{\mathcal{L}}(D, G) \subseteq C_{\mathcal{L}}(D, G)^{\perp}$.

2. If

$$2G - D = (\eta) \text{ and } \eta_{P_i}(1) = 1 \text{ for } i = 1, \dots, n,$$

then $C_{\mathcal{L}}(D, G)$ is self-dual.

Proof. 1. $2G - D \leq (\eta) \iff G \leq D - G + (\eta)$. Hence by Proposition 12.6

$$C_{\mathcal{L}}(D, G)^{\perp} = C_{\mathcal{L}}(D, D - G + (\eta)) \supseteq C_{\mathcal{L}}(D, G).$$

$$2. \ 2G - D = (\eta) \implies G = D - G + (\eta)$$

$$\implies C_{\mathcal{L}}(D, G)^{\perp} = C_{\mathcal{L}}(D, D - G + (\eta)) = C_{\mathcal{L}}(D, G).$$

□

Chapter 13

Rational AG Codes

Definition 13.1. An algebraic geometry code $C_{\mathcal{L}}(D, G)$ associated with divisors G and D of a rational function field $\mathbb{F}_q(z)/\mathbb{F}_q$ is said to be rational (as in Section 11 it is assumed that $D = P_1 + \dots + P_n$ with pairwise distinct places of degree one, and $\text{supp } G \cap \text{supp } D = \emptyset$).

Remark 13.2. The length of a rational AG code is bounded by $q + 1$ as $\mathbb{F}_q(z)$ has only $q + 1$ places of degree one: the pole P_∞ of z and for each $\alpha \in \mathbb{F}_q$, the zero P_α of $z - \alpha$.

Proposition 13.3 (Properties of Rational AG Codes). $C = C_{\mathcal{L}}(D, G)$ be a rational AG code over $\mathbb{F}_q$, and let n, k, d be the parameters of C . Then we have:

(a) $n \leq q + 1$.

(b) $k = 0 \iff \deg G < 0$, and $k = n \iff \deg G > n - 2$.

(c) For $0 \leq \deg G \leq n - 2$,

$$k = 1 + \deg G \quad \text{and} \quad d = n - \deg G.$$

In particular, C is an MDS code.

(d) $C^\perp$ is also a rational AG code.

Next we determine specific generator matrices for rational AG codes.

Proposition 13.4. *Let $C = C_{\mathcal{L}}(D, G)$ be a rational AG code over $\mathbb{F}_q$ with parameters n, k and d .*

- (a) *If $n \leq q$ then there exist pairwise distinct elements $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ and $v_1, \dots, v_n \in \mathbb{F}_q^\times$ (not necessarily distinct) such that*

$$C = \{(v_1 f(\alpha_1), v_2 f(\alpha_2), \dots, v_n f(\alpha_n)) \mid f \in \mathbb{F}_q[z] \text{ and } \deg f \leq k-1\}.$$

The matrix

$$M = \begin{pmatrix} v_1 & v_2 & \dots & v_n \\ \alpha_1 v_1 & \alpha_2 v_2 & \dots & \alpha_n v_n \\ \alpha_1^2 v_1 & \alpha_2^2 v_2 & \dots & \alpha_n^2 v_n \\ \vdots & \vdots & & \vdots \\ \alpha_1^{k-1} v_1 & \alpha_2^{k-1} v_2 & \dots & \alpha_n^{k-1} v_n \end{pmatrix}$$

is a generator matrix for C .

- (b) *If $n = q + 1$, C has a generator matrix*

$$M = \begin{pmatrix} v_1 & v_2 & \dots & v_{n-1} & 0 \\ \alpha_1 v_1 & \alpha_2 v_2 & \dots & \alpha_{n-1} v_{n-1} & 0 \\ \alpha_1^2 v_1 & \alpha_2^2 v_2 & \dots & \alpha_{n-1}^2 v_{n-1} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ \alpha_1^{k-1} v_1 & \alpha_2^{k-1} v_2 & \dots & \alpha_{n-1}^{k-1} v_{n-1} & 1 \end{pmatrix}$$

where $\mathbb{F}_q = \{\alpha_1, \dots, \alpha_{n-1}\}$ and $v_1, \dots, v_{n-1} \in \mathbb{F}_q^\times$.

Proof.

- (a) Let $D = P_1 + \dots + P_n$. As $n \leq q$, $\exists$ a place P of degree one which $\notin \text{Supp } D$. Choose a place $Q \neq P$ of degree one (e.g., $Q = P_1$). By Riemann-Roch Theorem 8.1, $\ell(Q - P) = 1$, hence $Q - P$ is a principal divisor (Corollary 5.9). Let $Q - P = (z)$. Hence z is a generating element of the rational function field over $\mathbb{F}_q$ and P is the pole divisor of z , hence $P = P_\infty$.

By Proposition 13.3 (c) assume $\deg G = k - 1 \geq 0$ (the case $k = 0$ being trivial). The divisor $(k - 1)P_\infty - G$ has degree zero, so it is principal say $(k - 1)P_\infty - G = (u)$ with $0 \neq u \in F$. The k elements $u, zu, \dots, z^{k-1}u$ are in $\mathcal{L}(G)$

and they are Linearly independent over $\mathbb{F}_q$. As $\ell(G) = k$, hence a basis for $\mathcal{L}(G)$ is

$$\mathcal{L}(G) = \{uf(z) \mid f \in \mathbb{F}_q[z] \text{ and } \deg f \leq k-1\}.$$

Let $\alpha_i := z(P_i)$ and $v_i := u(P_i)$

$$\implies (uf(z))(P_i) = u(P_i)f(z(P_i)) = v_i f(\alpha_i)$$

for $i = 1, \dots, n$. Hence

$$C = C_{\mathcal{L}}(D, G) = \{(v_1 f(\alpha_1), \dots, v_n f(\alpha_n)) \mid \deg f \leq k-1\}$$

The codeword in C corresponding to uz^j is $(v_1 \alpha_1^j, v_2 \alpha_2^j, \dots, v_n \alpha_n^j)$, so the given matrix is a generator matrix of C .

(b) Now we have $n = q + 1$ and we can choose z in such a way that $P_n = P_\infty$ is the pole of z . As above, $(k-1)P_\infty - G = (u)$ with $0 \neq u \in F$, and $\{u, zu, \dots, z^{k-1}u\}$ is a basis of $\mathcal{L}(G)$. For $1 \leq i \leq n-1 = q$ the elements $\alpha_i := z(P_i) \in \mathbb{F}_q$ are pairwise distinct, so $\mathbb{F}_q = \{\alpha_1, \dots, \alpha_{n-1}\}$. Moreover, $v_i := u(P_i) \in \mathbb{F}_q^\times$ for $i = 1, \dots, n-1$. For $0 \leq j \leq k-2$,

$$((uz^j)(P_1), \dots, (uz^j)(P_n)) = (\alpha_1^j v_1, \dots, \alpha_{n-1}^j v_{n-1}, 0)$$

but for $j = k-1$ holds

$$((uz^{k-1})(P_1), \dots, (uz^{k-1})(P_n)) = (\alpha_1^{k-1} v_1, \dots, \alpha_{n-1}^{k-1} v_{n-1}, \gamma)$$

with an element $0 \neq \gamma \in \mathbb{F}_q$. Substituting u by $\gamma^{-1}u$ gives the generator matrix. $\square$

Reed Solomon Code As AG Code

Definition 13.5. Let $\alpha = (\alpha_1, \dots, \alpha_n)$ where the α_i are distinct elements of $\mathbb{F}_q$, and let $v = (v_1, \dots, v_n)$ where the v_i are nonzero (not necessarily distinct) elements of $\mathbb{F}_q$. Then the Generalized Reed-Solomon code, denoted by $\text{GRS}_k(\alpha, v)$, consists of all vectors

$$(v_1 f(\alpha_1), \dots, v_n f(\alpha_n))$$

with $f(z) \in \mathbb{F}_q[z]$ and $\deg f \leq k-1$ (for a fixed $k \leq n$) In the case $\alpha = (b, b^2, \dots, b^n)$ (where $n = q-1$ and b is a primitive n -th root of unity) and $v = (1, 1, \dots, 1)$, $\text{GRS}_k(\alpha, v)$ is a Reed-Solomon code.

Obviously $\text{GRS}_k(\alpha, v)$ is an $[n, k]$ code, and Proposition 13.3(a) states that all rational AG codes over $\mathbb{F}_q$ of length $n \leq q$ are Generalized Reed Solomon codes. The converse is also true:

Proposition 13.6. *Every generalized Reed-Solomon code $\text{GRS}_k(\alpha, v)$ can be represented as a rational AG code.*

Proof. Let $\alpha = (\alpha_1, \dots, \alpha_n)$ with $\alpha_i \in \mathbb{F}_q$ and $v = (v_1, \dots, v_n)$ with $v_i \in \mathbb{F}_q^\times$. Consider the rational function field $F = \mathbb{F}_q(z)$. Denote by P_i the zero of $z - \alpha_i$ ($i = 1, \dots, n$) and by P_∞ the pole of z . Choose $u \in F$ such that

$$u(P_i) = v_i \quad \text{for} \quad i = 1, \dots, n. \quad (13.1)$$

Such an element exists by the Strong Approximation Theorem. (One can also determine a polynomial $u = u(z) \in \mathbb{F}_q[z]$ satisfying (13.1), by using Lagrange interpolation.) Now let

$$D := P_1 + \dots + P_n \quad \text{and} \quad G := (k-1)P_\infty - (u).$$

□

Chapter 14

Exercises- Algebraic Function Fields

Here are some solved questions from Exercises from Chapter 1 of [1].

[Q1.] Consider the rational function field $K(x)/K$ and a non-constant element $z = f(x)/g(x) \in K(x) \setminus K$, where $f(x), g(x) \in K[x]$ are relatively prime. We call $\deg(z) := \max\{\deg f(x), \deg g(x)\}$ the degree of z .

(i) Show that $[K(x) : K(z)] = \deg(z)$, and write down the minimal polynomial of x over $K(z)$.

(ii) Show that $K(x) = K(z)$ if and only if $z = (ax + b)/(cx + d)$ with $a, b, c, d \in K$ and $ad - bc \neq 0$.

Ans.(i)

$$\begin{aligned}
 [K(x) : K(z)] &= \deg(z_0) = \sum_P v_P(z) \deg P \\
 &= \begin{cases} P_{p(x)} v_{P_{p(x)}}(z) \cdot \deg P_{p(x)}, & p(x) | g(x) \text{ if } \deg(f(x)) \geq \deg(g(x)) \\ \sum_{P_{p(x)}} v_{P_{p(x)}}(z) \cdot \deg P_{p(x)} + v_{P_\infty} P_\infty & \text{if } \deg(f(x)) < \deg(g(x)) \end{cases} \\
 &= \begin{cases} \deg(f(x)) & \text{if } \deg(f(x)) > \deg(g(x)) \\ \deg(g(x)) & \text{if } \deg(f(x)) < \deg(g(x)) \end{cases}
 \end{aligned}$$

(ii) If $z = \frac{ax+b}{cx+d}$, $[K(x) : K(z)] = \deg(z) = 1 \implies K(x) = K(z)$.
 Conversely, let $K(x) = K(z) \implies \deg(z) = 1$ if $z = ax + b$ then

$\frac{x+b}{ax+b}(\in K(x)) \notin K(z)$. Similarly $z \neq 1/(ax+b)$. Hence z is of required form. $ad - bc \neq 0$ else z would be a constant.

[Q2.] For a field extension L/M we denote by $\text{Aut}(L/M)$ the group of automorphisms of L/M (i.e., automorphisms of L which are the identity on M). Let $K(x)/K$ be the rational function field over K . Show:

(i) For every $\sigma \in \text{Aut}(K(x)/K)$ there exist $a, b, c, d \in K$ such that $ad - bc \neq 0$ and $\sigma(x) = (ax + b)/(cx + d)$.

(ii) Given $a, b, c, d \in K$ with $ad - bc \neq 0$, there is a unique automorphism $\sigma \in \text{Aut}(K(x)/K)$ with $\sigma(x) = (ax + b)/(cx + d)$.

(iii) Denote by $\text{GL}_2(K)$ the group of invertible 2×2 matrices over K .

For $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \in \text{GL}_2(K)$ denote by σ_A the automorphism of $K(x)/K$ with $\sigma_A(x) = (ax + b)/(cx + d)$. Show that the map which sends A to σ_A , is a homomorphism from $\text{GL}_2(K)$ onto $\text{Aut}(K(x)/K)$. Its kernel is the set of diagonal matrices of the form $A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ with $a \in K^\times$, hence $\text{Aut}(K(x)/K) \cong \text{GL}_2(K)/K^\times$ (This group $\text{GL}_2(K)/K^\times$ is called the projective linear group and is denoted by $\text{PGL}_2(K)$.)

Ans. (i) As σ is automorphism, $K(x) = K(\sigma(x)) \iff \sigma(x) = (ax + b)/(cx + d)$ by Q1(ii).

(ii) As x generates field, every such $\sigma(x)$ defines unique automorphism.

(iii) define $\phi \begin{cases} \text{GL}_2(K) \rightarrow \text{Aut}(K(x)/K) \\ A \mapsto \sigma_A \end{cases}$

Now:-

$$\begin{aligned} & (AB) \\ &= \phi\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} A & C \\ B & D \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} aA + cB & aC + cD \\ bA + bD & bC + dD \end{bmatrix}\right) \\ &= \frac{(aA+cB)x+bA+Bd}{(aC+cD)x+bC+dD} \\ &= \frac{a(\frac{Ax+B}{Cx+D})+b}{c\frac{Ax+B}{Cx+D}+d} = \sigma_A \sigma_B \end{aligned}$$

Hence it is homomorphism.

[Q3.] If L is a field and G is a group of automorphisms of L , we denote by $L^G := \{w \in L \mid \sigma(w) = w \text{ for all } \sigma \in G\}$ the fixed field of G . It is well-known from algebra that if G is a finite group, then L/L^G is a finite extension of degree $[L : L^G] = \text{ord}(G)$. Now let $G \subseteq \text{Aut}(K(x)/K)$ be a finite subgroup of the automorphism group of the rational function field $K(x)$ over K , and put $z := \sum_{\sigma \in G} \sigma(x)$, $u := \prod_{\sigma \in G} \sigma(x)$. Show:

- (i) Either $z \in K$, or $K(z) = K(x)^G$.
- (ii) Either $u \in K$, or $K(u) = K(x)^G$.
- (iii) Find examples of finite subgroups $G \subseteq \text{Aut}(K(x)/K)$ for both alternatives in (i) (and also in (ii)).

Ans. (i) Note $\sigma(z) = \sum_{\sigma \in G} \sigma^2(z) = z \cdots (1)$. If $z \in K$ we are done. $K(z) = \{y = \sum a_i \frac{f_i(z)}{g_i(z)} \mid a_i \in K\} = \sigma(y) = \sum a_i \sigma(\frac{f_i(z)}{g_i(z)}) = \sum a_i \frac{f(\sigma(z))}{g(\sigma(z))} = z$ by (1)

(ii) Similar to (i).

(iii) An Example is finite subgroup $\langle \sigma_A \rangle$ generated from an idempotent matrix A as in Question Q2 (iii).

[Q4.] Let $K(x)$ be the rational function field over K . Find bases of the following Riemann-Roch spaces:

- (i) $\mathcal{L}(rP_\infty)$
- (ii) $\mathcal{L}(rP_\alpha)$
- (iii) $\mathcal{L}(P_{p(x)})$ where $r \geq 0$.

Ans. As in Example 5.13 $1, x, x^2, \dots, x^r$ forms basis for (i). Similarly for other parts (ii) and (iii) bases are $1, x - \alpha, (x - \alpha)^2, \dots, (x - \alpha)^r$ and $1, p(x), p(x)^2, \dots, p(x)^r$ respectively.

[Q5.] (*Representation of rational functions by partial fractions*) Show that every element $z \in K(x)$ can be written as:-

$$z = \sum_{i=1}^r \sum_{j=1}^{k_i} \frac{c_{ij}(x)}{p_i(x)^j} + h(x),$$

where

- (a) $p_1(x), \dots, p_r(x)$ are distinct monic irreducible polynomials in $K[x]$.
- (b) $k_1, \dots, k_r \geq 1$,
- (c) $c_{ij}(x) \in K[x]$ and $\deg(c_{ij}(x)) < \deg(p_i(x))$,
- (d) $c_{ik_i}(x) \neq 0$ for $1 \leq i \leq r$,
- (e) $h(x) \in K[x]$.

Ans. Let $z = f(x)/g(x)$.

We can write $f(x)$ as $f(x) = g(x)h(x) + r(x)$ by division algorithm $\implies f(x)/g(x) = h(x) + r(x)/g(x)$. We shall expand $r(x)/g(x)$.

Let $g(x) = g_1(x)g_2(x)$, where $g_i(x)$ are co-prime in $K[x]$. Let $r(x)d(x) = g_1q(x) + f_1$ by division algorithm, setting $f_2 = c(x)r(x) + q(x)g_2$.

$\implies \exists c(x), d(x)$ such that $c(x)g_1(x) + d(x)g_2(x) = 1 \implies r(x)/g(x) = r(x)(c(x)g_1(x) + d(x)g_2(x))/g_1(x)g_2(x) = r(x)c(x)/g_2(x) + r(x)d(x)/g_1(x) = (f_2 - q(x)g_2(x))/g_2(x) + (g_1q(x) + f_1)/g_1(x) = f_2(x)/g_2(x) + f_1(x)/g_1(x) + g_2(x), \deg f_i \leq g_1$.

Continuing similarly on $f_i(x)/g_i(x)$ until we get terms of the form $F(x)/g_i^k(x)$,

where $g_i(x)$ is irreducible. Now, derivative g'_i is co-prime to g_i , hence

$\exists c(x), d(x) \quad c(x)g'_i(x) + d(x)g_i(x) = 1 \implies F(x) = F(x)c(x)g'_i(x) + F(x)d(x)g_i(x) \cdots (1)$. By division algorithm $F(x)c(x)g'_i(x) = g(x)q(x) + f_k(x)$ and let $f_{k-1} = F(x)d_i(x) + q(x)$. Putting these in (1):-

$$F(x)/g_i^k(x) = (g_i(x)q(x) + f_k(x) + g_i(x)f_{k-1} - g_i(x)q(x))/g_i^k(x) = f_k(x)/g_i^k(x) + f_{k-1}(x)/g_i^{k-1}(x).$$

We can continue further for $f_{k-1}(x)/g_i^{k-1}(x)$ and get the required form.

[Q6.] Let $F = K(x)$ be the rational function field over K . Show directly that $A_F = A_F(0) + F$.

Ans. $A_F(0) := \{\alpha \in A_F \mid v_P(\alpha) \geq 0 \forall P\} = \{\alpha \in A_F \mid v_P(\alpha) \geq 0 \forall P\} = \{\alpha \in A_F \mid v_P(\alpha_P) \geq 0 \forall P\} = A_F$.

[Q7.] There are many analogies between algebraic function fields and algebraic number fields (i.e., finite extensions of the field $\mathbb{Q}$ of rational numbers). Here is a first example. A valuation ring of a field L is a subring $O \subsetneq L$ such that for all $z \in L$ one has that $z \in O$

or $z^{-1} \in O$.

(i) Show that every valuation ring is a local ring (i.e., it has a unique maximal ideal).

(ii) Now we consider the field $L = \mathbb{Q}$. Show that for every prime number $p \in \mathbb{Z}$, the set $\mathbb{Z}_{(p)} := \{a/b \in \mathbb{Q} \mid a, b \in \mathbb{Z} \text{ and } p \nmid b\}$ is a valuation ring of $\mathbb{Q}$. What is the maximal ideal of $\mathbb{Z}_{(p)}$? (iii) Let O be a valuation ring of $\mathbb{Q}$. Show that $O = \mathbb{Z}_{(p)}$ for some prime number p .

Ans.(i) Proof by Proposition 2.4 (a).

(ii) $\mathbb{Z}_{(p)} := \{a/b \in \mathbb{Q} \mid a, b \in \mathbb{Z} \text{ and } p \nmid b\}$

1. $a/b, p \nmid b \in \mathbb{Z}_p$.

2. $a/b, p \mid b, (a/b)^{-1} = b/a \in \mathbb{Z}_p$ as $p \nmid a$ as $\gcd(a, b) = 1$. Hence $\mathbb{Z}_{(p)}$ is a valuation ring. Maximal Ideal is set of non units, i.e., $\{a/b \in \mathbb{Q} \mid a, b \in \mathbb{Z} \text{ and } p \mid a\}$.

(iii) Let M is the unique maximal ideal of $O \implies M$ is a prime ideal $\implies M \cap \mathbb{Z} \implies M \cup \mathbb{Z} = p\mathbb{Z}$ for some prime number p . Then $pO = (M \cup \mathbb{Z})O \subseteq M$ as M is maximal. Let $a/b \in O$, $\gcd(a, b) = 1 \implies pa/b \in pO \subseteq M$, and $b/(pa) \notin O$. Assume that $p \mid b$, so that $\gcd(a, p) = 1$. It follows that $a \notin M \implies a^{-1} \in O \implies b/(pa) = ba^{-1}/p \in O$, a contradiction. Hence $p \nmid b \implies O \subseteq \mathbb{Z}_{(p)}$.

Now let $a/b \in \mathbb{Z}_{(p)}$, $(a, b) = 1 \implies b/a \notin p\mathbb{Z}_{(p)}$. Also, $M \subseteq M\mathbb{Z}_{(p)} \subseteq p\mathbb{Z}_{(p)}$, so $b/a \notin M \implies b/a$ is a unit in O and $a/b \in O$; otherwise $a/b \in O$ by the definition. Therefore $O = \mathbb{Z}_{(p)}$.

In the following exercises, F/K always denotes a function field of genus g with full constant field K .

[Q8.] Assume that $g > 0$ and A is a divisor with $\ell(A) > 0$. Show that $\ell(A) = \deg(A) + 1$ if and only if A is a principal divisor.

Ans. If A is principal, $i(A) = g$. Hence result holds by Reimann-Roch Theorem. Conversely from Clifford's Theorem, $1 + \dim(A) = \ell(A) \leq 1 + \frac{1}{2} \deg A \implies \dim A = 0$, hence A is principal.

[Q9.] Show that the following conditions are equivalent:

- (a) $g = 0$.
- (b) There is a divisor A with $\deg(A) = 2$ and $\ell(A) = 3$.
- (c) There is a divisor A with $\deg(A) \geq 1$ and $\ell(A) > \deg(A)$.
- (d) There is a divisor A with $\deg(A) \geq 1$ and $\ell(A) = \deg(A) + 1$.
In case of $\text{char } K = 2$, also the following condition is equivalent to the above:
- (e) There are elements $x, y \in F$ such that $F = K(x, y)$ and $y^2 = ax^2 + b$, with $a, b \in K^\times$.

Ans. (a) $\implies$ (b): $g = 0 \implies$ every canonical divisor W has degree $2g - 2 = -2$ (by corollary 8.2) $\implies \deg(-W) = 2 > 2g - 1 = -1$. Hence By Theorem 5.4 $\ell(-W) = \deg(-W) + 1 - g = 3$.

(b) $\implies$ (c): trivial.

(c) $\implies$ (d): We know $\ell(A) \geq \deg(A) + 1$. Hence by this inequality and (c) we get $\ell(A) = \deg(A) + 1$ and $\deg(A) \geq 1$.

(d) $\implies$ (a): Let $g > 0$. Hence by $\ell(A) = \deg(A) + 1$ and Q8., we get $\deg A = 0$, a contradiction. Hence $g = 0$

[Q10.] Let $\mathbb{R}(x)$ be the rational function field over the field of real numbers.

- (i) Show that the polynomial $f(T) := T^2 + (x^2 + 1) \in \mathbb{R}(x)[T]$ is irreducible over $\mathbb{R}(x)$. Let $F := \mathbb{R}(x, y)$, where $y^2 + x^2 + 1 = 0$. By (i), $[F : \mathbb{R}(x)] = 2$. Show:
- (ii) R is the full constant field of F , and $F/\mathbb{R}$ has genus $g = 0$.
- (iii) $F/\mathbb{R}$ is not a rational function field.
- (iv) All places of $F/\mathbb{R}$ have degree 2.

Ans. (i) Let f be reducible. Hence f has a zero in $\mathbb{R}(x) \implies T = \sqrt{-(x^2 + 1)} \notin \mathbb{R}(x)$, a contradiction. Hence f is irreducible.

(ii) As $[F : \mathbb{R}(x)] = [F : \mathbb{R}(y)] = 2 \implies x, y$ are transcendental and $\notin \overline{\mathbb{R}}$. Hence $\mathbb{R}$ is the field of constants by definition. F has genus 0 by Q9(e),

as $y^2 + x^2 + 1 = 0$ is given.

(iii) $[F : \mathbb{R}] = [F : \mathbb{R}(x)][\mathbb{R}(x) : \mathbb{R}] = 2 \neq 1$. Hence not a Rational Function Field.

(iv) For Any place P , $\deg P \leq [F : \mathbb{R}] = 2$. Now let Q' be a place of degree Q . As $g = 0$, $F/\mathbb{R}$ is a rational function field, a contradiction to (iii). Hence all places of $F/\mathbb{R}$ have degree 2.

[Q11.] Assume that $\text{char}(K) \neq 2$. Let $F = K(x, y)$ with

$$y^2 = f(x) \in K[x], \quad \deg f(x) = 2m + 1 \geq 3$$

Show:

- (i) K is the full constant field of F .
 - (ii) There is exactly one place $P \in \mathbb{P}_p$ which is a pole of x , and this place is also the only pole of y .
 - (iii) For every $s \geq 0$, the elements $1, x, x^2, \dots, x^r, y, xy, \dots, x^s y$ with $0 \leq s < r - m$ are in $\mathcal{O}(2rP)$.
 - (iv) The genus of F/K satisfies $g \leq m$.
- (i) Consider $y^2 - f(x) \in K(x)[y]$. It is an irreducible polynomial as $f(x)$ is of odd degree and can't be square of a polynomial. Hence as in Q10, x, y are transcendental over K hence K is the full constant field.

[Q13.] Assume that F/K has a place $P \in \mathbb{P}_F$ of degree one. Show that there exist $x, y \in F$ such that $[F : K(x)] = [F : K(y)] = 2g + 1$ and $F = K(x, y)$.

Ans. Choose given P and $n = 2g + 1$ in Proposition 8.10, we get x, y such that $[F : K(y)] = [F : K(x)] = \deg(x)_\infty = \deg nP = n = 2g + 1$. Now $x^n + y^n + 1 = 0$ is an irreducible polynomial over $R(x)$, $F = K(x, y)$.

[Q14.] Let V, W be vector spaces over K . A non-degenerate pairing of V and W is a bilinear map $s : V \times W \rightarrow K$ such that the following hold: For every $v \in V$ with $v \neq 0$ there is some $w \in W$ with $s(v, w) \neq 0$, and for every $w \in W$ with $w \neq 0$ there is some $v \in V$

with $s(v, w) = 0$. Now we consider a function field F/K , a divisor $A \in \text{Div}(F)$ and a non-zero Weil differential $\omega \in \Omega_F$. Let $W := (\omega)$. Show that the map $s : \mathcal{L}(W - A) \times A_F/(A_F(A) + F) \rightarrow K$ given by $s(x, \alpha) := \omega(x\alpha)$ is well-defined, and it is a non-degenerate pairing.

Ans. $\forall x$ we can choose adele $\alpha \neq 0$ such that $x\alpha \in A_F(A) + F$ such that $\omega(x\alpha) = 0$ by definition of x . Similarly for all α .

Bibliography

- [1] Henning Stichtenoth; *Algebraic Function Fields and Codes*, Springer, 2008.
- [2] P.B. Bhattacharya, S.K.Jain, S.R. Nagpaul; *Basic Abstract Algebra*, Cambridge University Press, Second Edition Press, 1995.
- [3] V. D. Goppa, *Codes On Algebraic Curves*, Soviet Math. Dokl. **24**, No.1, 1981, pp. 170-172.